

Notes on the Classical Theory of Linear Models

1 The Multiple Linear Regression Model

1.1 Model Specification and Matrix Notation

Definition 1.1 (Multiple Linear Regression Model). Given n observations, each consisting of a dependent variable Y_i and k independent variables $X_{i1}, \dots, X_{ik}$, the **multiple linear regression model** assumes

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n, \quad (1)$$

where β_0 is the intercept, $\beta_1, \dots, \beta_k$ are slope coefficients, and ε_i is the error term. Equivalently, in matrix form,

$$\boxed{\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}} \quad (2)$$

where

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}_{n \times 1}, \quad \mathbf{X} = \begin{pmatrix} 1 & X_{11} & \cdots & X_{1k} \\ 1 & X_{21} & \cdots & X_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \cdots & X_{nk} \end{pmatrix}_{n \times (k+1)}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}_{(k+1) \times 1}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}_{n \times 1}.$$

The matrix $\mathbf{X}$ is called the **design matrix**. Its first column is the all-ones vector $\mathbf{1} = (1, 1, \dots, 1)'$, corresponding to the intercept. We write $\mathbf{x}'_i$ for the i -th row of $\mathbf{X}$, so that $Y_i = \mathbf{x}'_i \boldsymbol{\beta} + \varepsilon_i$.

1.2 Classical Assumptions

Definition 1.2 (Classical Assumptions). The following assumptions are imposed on model (2):

(A1) **Linearity**: $E[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}$.

(A2) **Zero conditional mean**: $E[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$.

(A3) **Spherical errors**: $\text{var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$ for a constant $\sigma^2 > 0$. This comprises two sub-conditions:

- *Homoscedasticity*: $\text{var}(\varepsilon_i) = \sigma^2$ for all i (a constant).
- *No serial correlation*: $\text{cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$.

(A4) **Full rank**: $\text{rank}(\mathbf{X}) = k + 1$ (the columns of $\mathbf{X}$ are linearly independent). This requires $n \geq k + 1$. For variance estimation and inference (Sections 3–7 and 9), we additionally require $n > k + 1$ so that $\text{MSE} = \text{SSE}/(n - k - 1)$ (see Definition 3.1) is well-defined.

(A5) **Normality**: $\boldsymbol{\varepsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ (the multivariate normal distribution $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is defined in Definition 4.1, Section 4).

Assumptions (A1)–(A4) are called the **Gauss–Markov assumptions**. Assumption (A5) is additionally needed for exact distributional results in finite samples.

Remark 1.3. Throughout this article, we treat $\mathbf{X}$ as fixed (non-stochastic) and condition on it. All expectations, variances, and distributions are conditional on $\mathbf{X}$, but we suppress the conditioning notation for brevity.

Remark 1.4 (Relationship between (A1) and (A2)). Given the model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, assumptions (A1) and (A2) are logically equivalent: taking conditional expectations of both sides gives $E[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta} + E[\boldsymbol{\varepsilon} \mid \mathbf{X}]$, so $E[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}$ holds if and only if $E[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$. We state both because each form is convenient in different proofs: (A1) is closer to how the model is written, and (A2) is closer to how residuals are handled. Conditioning on $\mathbf{X}$ (previous remark) also means that inside any proof, $E[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ can be used without further notation—these are the conditional expectation and variance with the conditioning suppressed.

1.3 Prerequisites from Linear Algebra (assumed without proof)

The following standard facts are used below without proof; they are the only linear-algebra results imported from outside these notes. (The probabilistic prerequisites are listed in Section 4.1.) Notation: for $\mathbf{A} \in \mathbb{R}^{m \times p}$, $\mathcal{C}(\mathbf{A}) = \{\mathbf{A}\mathbf{v} : \mathbf{v} \in \mathbb{R}^p\} \subseteq \mathbb{R}^m$ is the **column space** of $\mathbf{A}$ and $\mathcal{N}(\mathbf{A}) = \{\mathbf{v} \in \mathbb{R}^p : \mathbf{A}\mathbf{v} = \mathbf{0}\}$ its **null space**; a symmetric matrix $\mathbf{A}$ is **positive semi-definite**, written $\mathbf{A} \succeq \mathbf{0}$, if $\mathbf{v}'\mathbf{A}\mathbf{v} \geq 0$ for all $\mathbf{v}$, and **positive definite** if $\mathbf{v}'\mathbf{A}\mathbf{v} > 0$ for all $\mathbf{v} \neq \mathbf{0}$. Vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^m$ are **orthogonal**, written $\mathbf{v} \perp \mathbf{w}$, if $\mathbf{v}'\mathbf{w} = 0$; for a subspace $\mathcal{S} \subseteq \mathbb{R}^m$, $\mathbf{v} \perp \mathcal{S}$ means $\mathbf{v} \perp \mathbf{w}$ for every $\mathbf{w} \in \mathcal{S}$, and $\mathcal{S}^\perp = \{\mathbf{v} \in \mathbb{R}^m : \mathbf{v} \perp \mathcal{S}\}$ is the **orthogonal complement** of $\mathcal{S}$; for a single vector $\mathbf{a}$ we write $\mathbf{a}^\perp = \mathcal{C}(\mathbf{a})^\perp = \{\mathbf{v} \in \mathbb{R}^m : \mathbf{a}'\mathbf{v} = 0\}$.

- (L1) *Rank-nullity.* For $\mathbf{A} \in \mathbb{R}^{m \times p}$, $\dim \mathcal{N}(\mathbf{A}) + \text{rank}(\mathbf{A}) = p$. Hence two matrices with the same number of columns and the same null space have the same rank. (Used in Theorem 2.4, Step 3.)
- (L2) *Trace.* $\text{tr}(\mathbf{A}\mathbf{B}) = \text{tr}(\mathbf{B}\mathbf{A})$ whenever both products are defined; in particular $\text{tr}(\mathbf{P}\mathbf{A}\mathbf{P}^{-1}) = \text{tr}(\mathbf{A})$. (Used in Lemma 2.8.)
- (L3) *Spectral theorem.* Every symmetric $\mathbf{A} \in \mathbb{R}^{n \times n}$ can be written $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ with $\mathbf{P}$ orthogonal ($\mathbf{P}'\mathbf{P} = \mathbf{P}\mathbf{P}' = \mathbf{I}$) and $\mathbf{\Lambda}$ diagonal, its entries being the eigenvalues of $\mathbf{A}$. (Used in Theorem 4.5.)
- (L4) *Square roots of positive definite matrices.* If $\mathbf{A}$ is symmetric positive definite, then $\mathbf{A} = \mathbf{P}\mathbf{A}\mathbf{P}'$ as in (L3) with all eigenvalues positive, and $\mathbf{A}^{1/2} := \mathbf{P}\mathbf{\Lambda}^{1/2}\mathbf{P}'$ is symmetric positive definite with $\mathbf{A}^{1/2}\mathbf{A}^{1/2} = \mathbf{A}$; its inverse $\mathbf{A}^{-1/2} := (\mathbf{A}^{1/2})^{-1}$ satisfies $\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{I}$. In particular $\mathbf{A} = \mathbf{L}\mathbf{L}'$ for some invertible $\mathbf{L}$ (for instance $\mathbf{L} = \mathbf{A}^{1/2}$, or the Cholesky factor). (Used in Theorems 7.4 and 7.10 and in the solutions to Exercises B4 and B5.)

2 Ordinary Least Squares

Standing assumption: from here to the end of these notes, assumption (A4) (full column rank of $\mathbf{X}$) is in force. It is what makes $(\mathbf{X}'\mathbf{X})^{-1}$ —and hence $\hat{\boldsymbol{\beta}}$, $\mathbf{H}$ and $\mathbf{M}$ —exist, and it is restated in a theorem only when the proof uses it in an additional way. Assumptions (A1)–(A3) and (A5) are invoked explicitly where needed; the results of this section use none of them.

2.1 Definition and the Normal Equations

Definition 2.1 (Ordinary Least Squares). The **OLS estimator** $\hat{\boldsymbol{\beta}}$ is the value of $\boldsymbol{\beta}$ that minimizes the sum of squared residuals:

$$Q(\boldsymbol{\beta}) = \sum_{i=1}^n (Y_i - \mathbf{x}_i'\boldsymbol{\beta})^2 = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2. \quad (3)$$

Lemma 2.2 (Matrix Differentiation Formulas). For $\boldsymbol{\beta} \in \mathbb{R}^p$, constant $\mathbf{a} \in \mathbb{R}^p$ and constant $\mathbf{A} \in \mathbb{R}^{p \times p}$, write $Df(\boldsymbol{\beta}) := \partial f / \partial \boldsymbol{\beta}'$ (numerator layout, row vector). Then

- (i) $D(\mathbf{a}'\boldsymbol{\beta}) = D(\boldsymbol{\beta}'\mathbf{a}) = \mathbf{a}'$;
- (ii) $D(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')$, which equals $2\boldsymbol{\beta}'\mathbf{A}$ when $\mathbf{A}' = \mathbf{A}$.

Proof. Both functions are polynomials in the coordinates $\beta_1, \dots, \beta_p$ of $\boldsymbol{\beta}$, so we compute the partial derivatives coordinate by coordinate; $Df(\boldsymbol{\beta})$ is the row vector $(\partial f / \partial \beta_1, \dots, \partial f / \partial \beta_p)$.

(i). $\mathbf{a}'\boldsymbol{\beta} = \sum_{i=1}^p a_i \beta_i$, so $\partial(\mathbf{a}'\boldsymbol{\beta}) / \partial \beta_j = a_j$ for each j ; assembling the row vector gives $D(\mathbf{a}'\boldsymbol{\beta}) = (a_1, \dots, a_p) = \mathbf{a}'$. The case $D(\boldsymbol{\beta}'\mathbf{a})$ follows from $\boldsymbol{\beta}'\mathbf{a} = \mathbf{a}'\boldsymbol{\beta}$.

(ii). $\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta} = \sum_{i=1}^p \sum_{l=1}^p \beta_i A_{il} \beta_l$. The coordinate β_j appears in the terms with $i = j$ and in the terms with $l = j$ (the term $i = l = j$ is $A_{jj}\beta_j^2$, whose derivative $2A_{jj}\beta_j$ is counted once in each group), so

$$\frac{\partial(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta})}{\partial \beta_j} = \sum_{l=1}^p A_{jl} \beta_l + \sum_{i=1}^p \beta_i A_{ij} = (\mathbf{A}\boldsymbol{\beta})_j + (\mathbf{A}'\boldsymbol{\beta})_j.$$

Hence $D(\beta' A \beta) = (A \beta + A' \beta)' = \beta' (A + A')$, which equals $2\beta' A$ when $A' = A$. $\square$

Remark 2.3 (The same formulas via differentials). Readers who know the differential calculus of Magnus and Neudecker (developed in the Matrix Calculus notes, and *not* assumed here) can obtain the same formulas from the linearity and product rule of d together with the *first identification theorem* $df(\beta) = c(\beta)' d\beta \iff Df(\beta) = c(\beta)'$: since $da = 0$ and $dA = 0$, $d(a' \beta) = a' d\beta$, and, using the scalar identity $(d\beta)' A \beta = \beta' A' d\beta$,

$$d(\beta' A \beta) = (d\beta)' A \beta + \beta' A d\beta = \beta' A' d\beta + \beta' A d\beta = \beta' (A + A') d\beta.$$

Nothing in these notes depends on this remark.

2.2 Derivation via Differentiation

Theorem 2.4 (OLS Estimator). Under assumption (A4), the OLS estimator is

$$\boxed{\hat{\beta} = (X'X)^{-1} X'Y.} \quad (4)$$

Proof. Step 1: Expand $Q(\beta)$.

$$\begin{aligned} Q(\beta) &= (Y - X\beta)'(Y - X\beta) \\ &= Y'Y - Y'X\beta - \beta'X'Y + \beta'X'X\beta \\ &= Y'Y - 2\beta'X'Y + \beta'X'X\beta. \end{aligned} \quad (5)$$

In the last line, $Y'X\beta$ is a scalar, hence equals its transpose $\beta'X'Y$.

Step 2: First-order condition. Apply Lemma 2.2 term by term, with $a = X'Y$ and $A = X'X$ (symmetric):

$$DQ(\beta) = -2(X'Y)' + 2\beta'X'X = -2Y'X + 2\beta'X'X.$$

Setting $DQ = 0'$ and transposing gives the **normal equation**

$$X'X\hat{\beta} = X'Y. \quad (6)$$

Step 3: Solve. By (A4), $\text{rank}(X) = k + 1$. We show $\text{rank}(X'X) = \text{rank}(X)$: $Xv = 0$ trivially implies $X'Xv = 0$; conversely, $X'Xv = 0$ gives $v'X'Xv = \|Xv\|^2 = 0$, hence $Xv = 0$. So X and $X'X$ have the same null space, hence the same rank. Thus $X'X$ is $(k + 1) \times (k + 1)$ of full rank, hence invertible. Multiplying (6) by $(X'X)^{-1}$:

$$\hat{\beta} = (X'X)^{-1} X'Y.$$

Step 4: Verify the second-order condition. The Hessian matrix is

$$\frac{\partial^2 Q}{\partial \beta \partial \beta'} = 2X'X.$$

Under assumption (A4), for any nonzero $v \in \mathbb{R}^{k+1}$, $Xv \neq 0$, so $v'X'Xv = \|Xv\|^2 > 0$. Hence $X'X$ is positive definite, so $\hat{\beta}$ is a local minimum. (Its inverse $(X'X)^{-1}$ is positive definite as well, a fact needed in Theorems 7.4 and 10.3: it is symmetric, being the inverse of a symmetric matrix, and for $v \neq 0$ the vector $w = (X'X)^{-1}v$ is nonzero (as $v = X'Xw$), so $v'(X'X)^{-1}v = w'X'Xw = \|Xw\|^2 > 0$.) Positive definiteness of the Hessian by itself certifies only a *local* minimum; that the minimum is global and unique follows from an exact expansion. For any $\beta \in \mathbb{R}^{k+1}$, write $Y - X\beta = (Y - X\hat{\beta}) - X(\beta - \hat{\beta})$; then

$$Q(\beta) = Q(\hat{\beta}) - 2(\beta - \hat{\beta})'X'(Y - X\hat{\beta}) + \|X(\beta - \hat{\beta})\|^2 = Q(\hat{\beta}) + \|X(\beta - \hat{\beta})\|^2 \geq Q(\hat{\beta}), \quad (7)$$

because $X'(Y - X\hat{\beta}) = 0$ is the normal equation (6). Equality holds iff $X(\beta - \hat{\beta}) = 0$, i.e. (by (A4)) iff $\beta = \hat{\beta}$. Thus $\hat{\beta}$ is the unique global minimizer of Q . $\square$

2.3 Derivation via Orthogonal Projection

Theorem 2.5 (OLS as Orthogonal Projection). The fitted vector $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$, that is, $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X})$ (the term is defined for a general subspace, in matrix form, in Remark 2.10), and the OLS estimator is $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$.

Proof. The OLS problem is to minimize $Q(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ over $\boldsymbol{\beta} \in \mathbb{R}^{k+1}$, equivalently, to find the point in $\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^{k+1}\}$ closest to $\mathbf{Y}$ in Euclidean norm.

Step 1: Orthogonality characterization of the minimum. We claim $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ minimizes $\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2$ over $\tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ iff $\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X})$.

($\Leftarrow$) Assume the orthogonality condition and let $\tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$. Since $\mathcal{C}(\mathbf{X})$ is a subspace, $\hat{\mathbf{Y}} - \tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$, so $(\mathbf{Y} - \hat{\mathbf{Y}}) \perp (\hat{\mathbf{Y}} - \tilde{\mathbf{Y}})$. Pythagoras applied to $\mathbf{Y} - \tilde{\mathbf{Y}} = (\mathbf{Y} - \hat{\mathbf{Y}}) + (\hat{\mathbf{Y}} - \tilde{\mathbf{Y}})$ gives

$$\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2 = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 + \|\hat{\mathbf{Y}} - \tilde{\mathbf{Y}}\|^2 \geq \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2,$$

with equality iff $\tilde{\mathbf{Y}} = \hat{\mathbf{Y}}$. So $\hat{\mathbf{Y}}$ is the unique minimizer.

($\Rightarrow$) Let $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ minimize. Suppose, for contradiction, some $\mathbf{v} \in \mathcal{C}(\mathbf{X})$ has $\mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}}) \neq 0$; normalize to $\|\mathbf{v}\|^2 = 1$. For $\tilde{\mathbf{Y}} = \hat{\mathbf{Y}} + t\mathbf{v} \in \mathcal{C}(\mathbf{X})$,

$$\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2 = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 - 2t\mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}}) + t^2,$$

minimized at $t^* = \mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}}) \neq 0$ with value $\|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 - (t^*)^2 < \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2$, contradicting optimality. Hence $\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X})$.

Step 2: Translate the orthogonality condition to a normal equation. The residual is orthogonal to every vector in $\mathcal{C}(\mathbf{X})$ if and only if it is orthogonal to every column of $\mathbf{X}$ (since those columns span $\mathcal{C}(\mathbf{X})$):

$$\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X}) \iff \mathbf{X}'(\mathbf{Y} - \hat{\mathbf{Y}}) = \mathbf{0}.$$

Writing $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ for some $\hat{\boldsymbol{\beta}}$, this becomes $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}$, the same normal equation (6) obtained by differentiation.

Step 3: Solve. Under assumption (A4), $\mathbf{X}'\mathbf{X}$ is invertible, so $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$, and the minimizer $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ exists and is unique. $\square$

2.4 Hat Matrix, Residual Maker, and Idempotent Matrices

Definition 2.6 (Hat Matrix and Residual Maker Matrix). The **hat matrix** is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'. \quad (8)$$

The **residual maker matrix** (or **annihilator matrix**) is

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}. \quad (9)$$

Then $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{Y}$, and the **residual vector** $\mathbf{e} := \mathbf{Y} - \hat{\mathbf{Y}}$, with components $e_i = Y_i - \hat{Y}_i$, satisfies $\mathbf{e} = \mathbf{Y} - \mathbf{H}\mathbf{Y} = \mathbf{M}\mathbf{Y}$.

Definition 2.7 (Idempotent Matrix). A square matrix $\mathbf{A}$ is **idempotent** if $\mathbf{A}^2 = \mathbf{A}$.

Lemma 2.8 (Trace Equals Rank for Idempotent Matrices). If $\mathbf{A}$ is an $n \times n$ idempotent matrix, then $\text{tr}(\mathbf{A}) = \text{rank}(\mathbf{A})$.

Proof. We show every idempotent matrix is diagonalizable with eigenvalues in $\{0, 1\}$.

Step 1: Eigenvalues are in $\{0, 1\}$. If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $\lambda\mathbf{v} = \mathbf{A}\mathbf{v} = \mathbf{A}^2\mathbf{v} = \lambda^2\mathbf{v}$, forcing $\lambda \in \{0, 1\}$.

Step 2: $\mathbb{R}^n = \mathcal{N}(\mathbf{A}) \oplus \mathcal{C}(\mathbf{A})$. Any $\mathbf{v} \in \mathbb{R}^n$ decomposes as $\mathbf{v} = (\mathbf{v} - \mathbf{A}\mathbf{v}) + \mathbf{A}\mathbf{v}$, with $\mathbf{A}\mathbf{v} \in \mathcal{C}(\mathbf{A})$ and $\mathbf{v} - \mathbf{A}\mathbf{v} \in \mathcal{N}(\mathbf{A})$ (since $\mathbf{A}(\mathbf{v} - \mathbf{A}\mathbf{v}) = \mathbf{A}\mathbf{v} - \mathbf{A}^2\mathbf{v} = \mathbf{0}$). The sum is direct: if $\mathbf{v} = \mathbf{A}\mathbf{w} \in \mathcal{C}(\mathbf{A}) \cap \mathcal{N}(\mathbf{A})$, then $\mathbf{v} = \mathbf{A}\mathbf{w} = \mathbf{A}^2\mathbf{w} = \mathbf{A}\mathbf{v} = \mathbf{0}$.

Step 3: Diagonalizability. Any $\mathbf{u} \in \mathcal{C}(\mathbf{A})$ satisfies $\mathbf{A}\mathbf{u} = \mathbf{u}$ ($\mathbf{u} = \mathbf{A}\mathbf{w}$, so $\mathbf{A}\mathbf{u} = \mathbf{A}^2\mathbf{w} = \mathbf{A}\mathbf{w} = \mathbf{u}$). Combining a basis of $\mathcal{N}(\mathbf{A})$ (eigenvalue 0) and a basis of $\mathcal{C}(\mathbf{A})$ (eigenvalue 1) gives an eigenvector basis of $\mathbb{R}^n$ by Step 2. Thus $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$ with $\mathbf{\Lambda}$ diagonal and entries in $\{0, 1\}$.

Step 4: Conclude. Let $r = \text{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$. Then $\mathbf{\Lambda}$ has r ones and $n - r$ zeros, so by the cyclic property of trace ((L2)),

$$\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}) = \text{tr}(\mathbf{\Lambda}) = r = \text{rank}(\mathbf{A}). \quad \square$$

Theorem 2.9 (Properties of $\mathbf{H}$ and $\mathbf{M}$). The matrices $\mathbf{H}$ and $\mathbf{M}$ satisfy:

- (i) $\mathbf{H}$ and $\mathbf{M}$ are both symmetric and idempotent.
- (ii) $\mathbf{H}\mathbf{X} = \mathbf{X}$ and $\mathbf{M}\mathbf{X} = \mathbf{0}$.
- (iii) $\mathbf{H}\mathbf{M} = \mathbf{M}\mathbf{H} = \mathbf{0}$.
- (iv) $\text{tr}(\mathbf{H}) = k + 1$ and $\text{tr}(\mathbf{M}) = n - k - 1$.

Proof. (i) **Symmetry of $\mathbf{H}$:** Since $\mathbf{X}'\mathbf{X}$ is symmetric, so is $(\mathbf{X}'\mathbf{X})^{-1}$:

$$\mathbf{H}' = [\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']' = \mathbf{X}[(\mathbf{X}'\mathbf{X})^{-1}]'\mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{H}.$$

Idempotency of $\mathbf{H}$:

$$\mathbf{H}^2 = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \underbrace{\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}}_{\mathbf{I}_{k+1}} \mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{H}.$$

Symmetry and idempotency of $\mathbf{M}$: $\mathbf{M}' = (\mathbf{I} - \mathbf{H})' = \mathbf{I} - \mathbf{H}' = \mathbf{I} - \mathbf{H} = \mathbf{M}$. Also, $\mathbf{M}^2 = (\mathbf{I} - \mathbf{H})^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H}^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H} = \mathbf{I} - \mathbf{H} = \mathbf{M}$.

(ii) $\mathbf{H}\mathbf{X} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{X}\mathbf{I}_{k+1} = \mathbf{X}$, and $\mathbf{M}\mathbf{X} = (\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{X} - \mathbf{X} = \mathbf{0}$.

(iii) $\mathbf{H}\mathbf{M} = \mathbf{H}(\mathbf{I} - \mathbf{H}) = \mathbf{H} - \mathbf{H}^2 = \mathbf{H} - \mathbf{H} = \mathbf{0}$, and $\mathbf{M}\mathbf{H} = (\mathbf{I} - \mathbf{H})\mathbf{H} = \mathbf{H} - \mathbf{H}^2 = \mathbf{0}$.

(iv) By Lemma 2.8, $\text{tr}(\mathbf{H}) = \text{rank}(\mathbf{H})$. We show $\text{rank}(\mathbf{H}) = k + 1$ in two steps.

Step 1: $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$. The inclusion $\mathcal{C}(\mathbf{H}) \subseteq \mathcal{C}(\mathbf{X})$ is immediate: every column of $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is of the form $\mathbf{X}\mathbf{v}$ for some $\mathbf{v} \in \mathbb{R}^{k+1}$. Conversely, $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{H})$ because $\mathbf{H}\mathbf{X} = \mathbf{X}$ by (ii), so every column of $\mathbf{X}$ equals $\mathbf{H}$ times a column of $\mathbf{X}$, hence lies in $\mathcal{C}(\mathbf{H})$.

Step 2: $\dim \mathcal{C}(\mathbf{X}) = k + 1$ by assumption (A4).

Combining, $\text{rank}(\mathbf{H}) = \dim \mathcal{C}(\mathbf{H}) = \dim \mathcal{C}(\mathbf{X}) = k + 1$, so $\text{tr}(\mathbf{H}) = k + 1$. Finally, $\text{tr}(\mathbf{M}) = \text{tr}(\mathbf{I}) - \text{tr}(\mathbf{H}) = n - (k + 1) = n - k - 1$. $\square$

Remark 2.10 ($\mathbf{H}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$). For a subspace $\mathcal{S} \subseteq \mathbb{R}^n$, a matrix $\mathbf{P}$ is called the **orthogonal projection onto $\mathcal{S}$** if $\mathbf{P}\mathbf{v} \in \mathcal{S}$ and $\mathbf{v} - \mathbf{P}\mathbf{v} \perp \mathcal{S}$ for every $\mathbf{v} \in \mathbb{R}^n$. By Step 1 of the proof of Theorem 2.5 (which used only that $\mathcal{C}(\mathbf{X})$ is a subspace), $\mathbf{P}\mathbf{v}$ is then the unique point of $\mathcal{S}$ closest to $\mathbf{v}$; hence there is at most one such matrix. One always exists: if $\mathcal{S} = \{\mathbf{0}\}$, the zero matrix is the orthogonal projection onto $\mathcal{S}$ (it maps every $\mathbf{v}$ to $\mathbf{0} \in \mathcal{S}$, and $\mathbf{v} - \mathbf{0} = \mathbf{v} \perp \{\mathbf{0}\}$ since $\mathbf{v}'\mathbf{0} = 0$), and it is symmetric and idempotent with column space $\{\mathbf{0}\} = \mathcal{S}$; otherwise, writing $\mathcal{S} = \mathcal{C}(\mathbf{B})$ with the columns of $\mathbf{B}$ a basis of $\mathcal{S}$ (so that $\mathbf{B}$ has full column rank), Definition 2.6 and Theorem 2.9 apply with $\mathbf{B}$ in place of $\mathbf{X}$, and $\mathbf{B}(\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'$ is the orthogonal projection onto $\mathcal{S}$ (it maps $\mathbf{v}$ into $\mathcal{C}(\mathbf{B})$,

and $\mathbf{B}'[\mathbf{v} - \mathbf{B}(\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'\mathbf{v}] = \mathbf{0}$); it is symmetric and idempotent with column space $\mathcal{S}$. In particular $\mathbf{H}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$: $\mathbf{H}\mathbf{v} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{X}'(\mathbf{v} - \mathbf{H}\mathbf{v}) = \mathbf{X}'\mathbf{M}\mathbf{v} = \mathbf{0}$ by (ii).

Combining (i), (ii), and the identity $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$ proved above: $\mathbf{H}$ is symmetric, idempotent, and has column space $\mathcal{C}(\mathbf{X})$. These three properties characterize the orthogonal projection onto $\mathcal{C}(\mathbf{X})$, in the sense that any symmetric idempotent $\mathbf{P}$ with $\mathcal{C}(\mathbf{P}) = \mathcal{C}(\mathbf{X})$ equals $\mathbf{H}$. Indeed, for $\mathbf{v} \in \mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{P})$, say $\mathbf{v} = \mathbf{P}\mathbf{w}$, we get $\mathbf{P}\mathbf{v} = \mathbf{P}^2\mathbf{w} = \mathbf{P}\mathbf{w} = \mathbf{v}$; for $\mathbf{v} \perp \mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{P})$, we get $\|\mathbf{P}\mathbf{v}\|^2 = \mathbf{v}'\mathbf{P}'\mathbf{P}\mathbf{v} = \mathbf{v}'(\mathbf{P}\mathbf{v}) = 0$ because $\mathbf{P}\mathbf{v} \in \mathcal{C}(\mathbf{P})$; and every $\mathbf{v} \in \mathbb{R}^n$ decomposes as $\mathbf{v} = \mathbf{H}\mathbf{v} + \mathbf{M}\mathbf{v}$ with $\mathbf{H}\mathbf{v} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{M}\mathbf{v} \perp \mathcal{C}(\mathbf{X})$, so $\mathbf{P}\mathbf{v} = \mathbf{P}(\mathbf{H}\mathbf{v}) + \mathbf{P}(\mathbf{M}\mathbf{v}) = \mathbf{H}\mathbf{v} + \mathbf{0} = \mathbf{H}\mathbf{v}$. Geometrically, $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ is the closest point in $\mathcal{C}(\mathbf{X})$ to $\mathbf{Y}$ in Euclidean distance (this is the content of §2, specifically the orthogonal projection derivation). Analogously, $\mathbf{M} = \mathbf{I} - \mathbf{H}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})^\perp$: $\mathbf{M}\mathbf{v} \perp \mathcal{C}(\mathbf{X})$ as just noted, and $\mathbf{v} - \mathbf{M}\mathbf{v} = \mathbf{H}\mathbf{v} \in \mathcal{C}(\mathbf{X})$ is orthogonal to $\mathcal{C}(\mathbf{X})^\perp$.

2.5 Errors Versus Residuals

Remark 2.11 (Errors versus residuals). The **error** $\varepsilon_i = Y_i - \mathbf{x}_i'\boldsymbol{\beta}$ is the deviation from the true (unknown) regression function; it is *unobservable*. The **residual** $e_i = Y_i - \mathbf{x}_i'\hat{\boldsymbol{\beta}} = Y_i - \hat{Y}_i$ is the deviation from the fitted regression function; it is *observable*. In matrix form, $\mathbf{e} = \mathbf{M}\mathbf{Y} = \mathbf{M}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \mathbf{M}\boldsymbol{\varepsilon}$ (since $\mathbf{M}\mathbf{X} = \mathbf{0}$ by Theorem 2.9(ii)). The residual vector is a linear transformation of the error vector.

2.6 Algebraic Properties of OLS Residuals

Theorem 2.12 (Implications of the Normal Equations). The OLS residuals satisfy:

- (i) $\mathbf{X}'\mathbf{e} = \mathbf{0}$ (residuals are orthogonal to every column of $\mathbf{X}$).
- (ii) $\sum_{i=1}^n e_i = 0$ (residuals sum to zero), provided the model includes an intercept.
- (iii) $\hat{\mathbf{Y}}'\mathbf{e} = 0$ (residuals are orthogonal to fitted values).

Proof. (i) From the normal equation $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}$, rearranging:

$$\mathbf{X}'\mathbf{Y} - \mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0} \iff \mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0} \iff \mathbf{X}'\mathbf{e} = \mathbf{0}.$$

(ii) The first column of $\mathbf{X}$ is $\mathbf{1}$. The first component of the vector equation $\mathbf{X}'\mathbf{e} = \mathbf{0}$ is $\mathbf{1}'\mathbf{e} = \sum_{i=1}^n e_i = 0$. This relies on the presence of the intercept column.

(iii) Since $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$:

$$\hat{\mathbf{Y}}'\mathbf{e} = (\mathbf{X}\hat{\boldsymbol{\beta}})'\mathbf{e} = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{e} = \hat{\boldsymbol{\beta}}' \cdot \mathbf{0} = 0.$$

Geometrically, $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{e} \perp \mathcal{C}(\mathbf{X})$. □

2.7 Leverages

Definition 2.13 (Leverage). The **leverage** of the i -th observation is h_{ii} , the i -th diagonal entry of $\mathbf{H}$.

Theorem 2.14 (Properties of Leverages).

- (i) $0 \leq h_{ii} \leq 1$ for all i .
- (ii) $\sum_{i=1}^n h_{ii} = k + 1$.

Proof. (i) Since $\mathbf{H}$ is symmetric and idempotent, $\mathbf{H} = \mathbf{H}^2 = \mathbf{H}'\mathbf{H}$. Comparing the (i, i) entry on both sides using $h_{ji} = h_{ij}$:

$$h_{ii} = \sum_{j=1}^n h_{ij}h_{ji} = \sum_{j=1}^n h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2.$$

The rightmost sum is ≥ 0 , so $h_{ii} \geq h_{ii}^2$, which rearranges to $h_{ii}(1 - h_{ii}) \geq 0$. The leftmost expression $h_{ii} = \sum_j h_{ij}^2 \geq 0$ is also a sum of squares. Combining $h_{ii} \geq 0$ with $h_{ii}(1 - h_{ii}) \geq 0$ forces $h_{ii} \leq 1$.

(ii) $\sum_{i=1}^n h_{ii} = \text{tr}(\mathbf{H}) = k + 1$ by Theorem 2.9(iv). $\square$

3 Finite-Sample Properties Without Normality

In this section, only assumptions (A1)–(A4) are used. No distributional assumption on $\boldsymbol{\varepsilon}$ is needed.

Before deriving the finite-sample properties of $\hat{\boldsymbol{\beta}}$, we collect the notation for the sum-of-squares and mean-square quantities used throughout the remaining sections.

Definition 3.1 (Sums of Squares and Mean Squares). Given the fitted model $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$, define:

- **Total sum of squares:** $\text{SST} = \sum_{i=1}^n (Y_i - \bar{Y})^2$, where $\bar{Y} = n^{-1} \sum_i Y_i$.
- **Regression sum of squares:** $\text{SSR} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$.
- **Error (residual) sum of squares:** $\text{SSE} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \mathbf{e}'\mathbf{e}$, where $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$ is the residual vector of Definition 2.6.
- **Mean squared error:** $\text{MSE} = s^2 = \text{SSE}/(n - k - 1)$, defined when $n > k + 1$.
- **Mean squared regression:** $\text{MSR} = \text{SSR}/k$, defined when $k \geq 1$ (at least one non-intercept regressor).

The decomposition $\text{SST} = \text{SSR} + \text{SSE}$ is proved in Theorem 6.4 (§6); the unbiasedness $\text{E}[\text{MSE}] = \sigma^2$ is proved in Theorem 3.7 below.

We also record a tool used repeatedly in this and later sections.

Lemma 3.2 (Variance and covariance under linear transformations). Let $\mathbf{z}$ be an $n \times 1$ random vector with mean $\boldsymbol{\mu} = \text{E}[\mathbf{z}]$ and covariance matrix $\text{var}(\mathbf{z}) = \text{E}[(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})']$. For two random vectors $\mathbf{U}$ ($m \times 1$) and $\mathbf{V}$ ($l \times 1$) with finite second moments, the **cross-covariance matrix** is the $m \times l$ matrix

$$\text{cov}(\mathbf{U}, \mathbf{V}) = \text{E}[(\mathbf{U} - \text{E}[\mathbf{U}])(\mathbf{V} - \text{E}[\mathbf{V}])'],$$

so that $\text{var}(\mathbf{z}) = \text{cov}(\mathbf{z}, \mathbf{z})$. Let $\mathbf{A}$ ($m \times n$) and $\mathbf{B}$ ($l \times n$) be constant matrices and $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{d} \in \mathbb{R}^l$ constant vectors. Then

$$(i) \quad \text{E}[\mathbf{Az} + \mathbf{b}] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b} \quad \text{and} \quad \text{var}(\mathbf{Az} + \mathbf{b}) = \mathbf{A} \text{var}(\mathbf{z}) \mathbf{A}'; \quad (10)$$

$$(ii) \quad \text{cov}(\mathbf{Az} + \mathbf{b}, \mathbf{Bz} + \mathbf{d}) = \mathbf{A} \text{var}(\mathbf{z}) \mathbf{B}'.$$

In particular, under assumption (A3), $\text{var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$.

Proof. (i) Since $\mathbf{A}$ and $\mathbf{b}$ are constant, $\text{E}[\mathbf{Az} + \mathbf{b}] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$. Therefore

$$\begin{aligned} \text{var}(\mathbf{Az} + \mathbf{b}) &= \text{E}[(\mathbf{Az} + \mathbf{b} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})(\mathbf{Az} + \mathbf{b} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})'] \\ &= \text{E}[\mathbf{A}(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})' \mathbf{A}'] \\ &= \mathbf{A} \text{E}[(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})'] \mathbf{A}' = \mathbf{A} \text{var}(\mathbf{z}) \mathbf{A}', \end{aligned}$$

since constant matrices pull outside the expectation.

(ii) By the first part of (i), $\mathbf{Az} + \mathbf{b} - \text{E}[\mathbf{Az} + \mathbf{b}] = \mathbf{A}(\mathbf{z} - \boldsymbol{\mu})$ and $\mathbf{Bz} + \mathbf{d} - \text{E}[\mathbf{Bz} + \mathbf{d}] = \mathbf{B}(\mathbf{z} - \boldsymbol{\mu})$, so

$$\text{cov}(\mathbf{Az} + \mathbf{b}, \mathbf{Bz} + \mathbf{d}) = \text{E}[\mathbf{A}(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})' \mathbf{B}'] = \mathbf{A} \text{E}[(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})'] \mathbf{B}' = \mathbf{A} \text{var}(\mathbf{z}) \mathbf{B}'.$$

(Taking $\mathbf{B} = \mathbf{A}$, $\mathbf{d} = \mathbf{b}$ recovers (10).)

Finally, applying (i) to $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ (with $\mathbf{A} = \mathbf{I}_n$, $\mathbf{b} = \mathbf{X}\boldsymbol{\beta}$ constant given the conditioning on $\mathbf{X}$) gives $\text{var}(\mathbf{Y}) = \text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$ by (A3). $\square$

3.1 Unbiasedness

Theorem 3.3 (Unbiasedness of $\hat{\beta}$). Under (A2) and (A4) (the latter is needed for $\hat{\beta}$ to exist), the OLS estimator is unbiased:

$$\boxed{\mathbb{E}[\hat{\beta}] = \beta.} \quad (11)$$

Proof. Substituting $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$ into $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$:

$$\begin{aligned} \hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \varepsilon) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon \\ &= \mathbf{I}_{k+1}\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon \\ &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon. \end{aligned} \quad (12)$$

Taking expectations, using (A2) ($\mathbb{E}[\varepsilon] = \mathbf{0}$):

$$\mathbb{E}[\hat{\beta}] = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}[\varepsilon] = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \cdot \mathbf{0} = \beta. \quad \square$$

3.2 Variance–Covariance Matrix

Theorem 3.4 (Variance–Covariance Matrix of $\hat{\beta}$). Under (A3) and (A4):

$$\boxed{\text{var}(\hat{\beta}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.} \quad (13)$$

Proof. From (12), $\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon$, which is a linear transformation of ε (with a constant shift). Applying Lemma 3.2 with $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$:

$$\text{var}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \text{var}(\varepsilon) \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}.$$

By (A3), $\text{var}(\varepsilon) = \sigma^2\mathbf{I}_n$. Substituting:

$$\text{var}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1},$$

using $\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{I}_{k+1}$. The diagonal entry $[\text{var}(\hat{\beta})]_{jj} = \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$ gives $\text{var}(\hat{\beta}_j)$. $\square$

3.3 The Gauss–Markov Theorem

Theorem 3.5 (Gauss–Markov Theorem). Under (A1)–(A4), the OLS estimator $\hat{\beta}$ is the **Best Linear Unbiased Estimator** (BLUE) of β . That is, for any other linear unbiased estimator $\tilde{\beta} = \mathbf{C}\mathbf{Y}$ (where $\mathbf{C}$ is a $(k+1) \times n$ constant matrix), the difference $\text{var}(\tilde{\beta}) - \text{var}(\hat{\beta})$ is positive semi-definite. In particular, $\text{var}(\tilde{\beta}_j) \geq \text{var}(\hat{\beta}_j)$ for every j .

Proof. Step 1: Necessary condition for unbiasedness. By (A1), $\mathbb{E}[\tilde{\beta}] = \mathbf{C}\mathbf{X}\beta = \beta$ for all $\beta \in \mathbb{R}^{k+1}$ forces $\mathbf{C}\mathbf{X} = \mathbf{I}_{k+1}$ (setting $\mathbf{v} = \boldsymbol{\iota}_j$ in $\mathbf{C}\mathbf{X}\mathbf{v} = \mathbf{v}$ reads off the j -th column; here and below $\boldsymbol{\iota}_j$ denotes the j -th standard basis vector, i.e. the j -th column of the identity matrix of the relevant dimension).

Step 2: Decompose $\mathbf{C}$. Write $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}$, where $\mathbf{D}$ is the excess beyond the OLS coefficient matrix. From $\mathbf{C}\mathbf{X} = \mathbf{I}_{k+1}$:

$$\mathbf{D}\mathbf{X} = \mathbf{C}\mathbf{X} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{I}_{k+1} - \mathbf{I}_{k+1} = \mathbf{0}.$$

Step 3: Compute $\text{var}(\tilde{\beta})$. Since $\tilde{\beta} = \mathbf{C}\mathbf{Y}$ with $\mathbf{C}$ constant, Lemma 3.2 gives

$$\text{var}(\tilde{\beta}) = \mathbf{C} \text{var}(\mathbf{Y}) \mathbf{C}' = \sigma^2 \mathbf{C} \mathbf{C}' = \sigma^2 [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}] [\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}'].$$

Expanding yields four terms:

$$\begin{aligned}
\sigma^2 \mathbf{C}\mathbf{C}' &= \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{X} (\mathbf{X}'\mathbf{X})^{-1} && \text{(term A)} \\
&+ \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{D}' && \text{(term B)} \\
&+ \sigma^2 \mathbf{D}\mathbf{X} (\mathbf{X}'\mathbf{X})^{-1} && \text{(term C)} \\
&+ \sigma^2 \mathbf{D}\mathbf{D}'. && \text{(term D)}
\end{aligned}$$

Term A simplifies to $\sigma^2 (\mathbf{X}'\mathbf{X})^{-1}$ because $\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{I}_{k+1}$. Term C vanishes because $\mathbf{D}\mathbf{X} = \mathbf{0}$ (Step 2). Term B equals (term C)', so it also vanishes. Hence:

$$\text{var}(\tilde{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} + \sigma^2 \mathbf{D}\mathbf{D}' = \text{var}(\hat{\boldsymbol{\beta}}) + \sigma^2 \mathbf{D}\mathbf{D}'.$$

Since $\mathbf{D}\mathbf{D}'$ is positive semi-definite (for any $\mathbf{v} \in \mathbb{R}^{k+1}$, $\mathbf{v}'\mathbf{D}\mathbf{D}'\mathbf{v} = \|\mathbf{D}'\mathbf{v}\|^2 \geq 0$):

$$\text{var}(\tilde{\boldsymbol{\beta}}) - \text{var}(\hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{D}\mathbf{D}' \succeq \mathbf{0}.$$

Equality holds if and only if $\mathbf{D}'\mathbf{v} = \mathbf{0}$ for all $\mathbf{v}$, i.e., $\mathbf{D} = \mathbf{0}$; equivalently, $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$. $\square$

3.4 Estimation of σ^2

Lemma 3.6 (Expected Value of a Quadratic Form). Let $\mathbf{z}$ be an $n \times 1$ random vector with $\mathbb{E}[\mathbf{z}] = \boldsymbol{\mu}$ and $\text{var}(\mathbf{z}) = \boldsymbol{\Sigma}$. Let $\mathbf{A}$ be an $n \times n$ constant matrix. Then:

$$\mathbb{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \text{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}. \quad (14)$$

In particular, if $\mathbb{E}[\mathbf{z}] = \mathbf{0}$, then $\mathbb{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$.

Proof. The quadratic form is a scalar: $\mathbf{z}'\mathbf{A}\mathbf{z} = \sum_{i,j} a_{ij} z_i z_j$. Taking expectations:

$$\mathbb{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \sum_{i,j} a_{ij} \mathbb{E}[z_i z_j].$$

From $\text{cov}(z_i, z_j) = \mathbb{E}[z_i z_j] - \mu_i \mu_j$, we get $\mathbb{E}[z_i z_j] = \Sigma_{ij} + \mu_i \mu_j$. Substituting:

$$\mathbb{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \sum_{i,j} a_{ij} \Sigma_{ij} + \sum_{i,j} a_{ij} \mu_i \mu_j.$$

The first sum equals $\text{tr}(\mathbf{A}\boldsymbol{\Sigma})$: indeed, $\text{tr}(\mathbf{A}\boldsymbol{\Sigma}) = \sum_i (\mathbf{A}\boldsymbol{\Sigma})_{ii} = \sum_i \sum_j a_{ij} \Sigma_{ji} = \sum_{i,j} a_{ij} \Sigma_{ij}$, where the last step uses the symmetry $\Sigma_{ji} = \Sigma_{ij}$. The second sum equals $\boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$: indeed, $\boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu} = \sum_i \mu_i (\mathbf{A}\boldsymbol{\mu})_i = \sum_i \mu_i \sum_j a_{ij} \mu_j = \sum_{i,j} a_{ij} \mu_i \mu_j$. Combining yields (14). $\square$

Theorem 3.7 (Unbiasedness of MSE). Under (A2)–(A4), the mean squared error

$$s^2 = \text{MSE} = \frac{\text{SSE}}{n - k - 1} = \frac{\mathbf{e}'\mathbf{e}}{n - k - 1} \quad (15)$$

is an unbiased estimator of σ^2 : $\mathbb{E}[\text{MSE}] = \sigma^2$.

Proof. Step 1: Express SSE as a quadratic form in $\boldsymbol{\varepsilon}$. Since $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ (Remark 2.11):

$$\text{SSE} = \mathbf{e}'\mathbf{e} = (\mathbf{M}\boldsymbol{\varepsilon})'(\mathbf{M}\boldsymbol{\varepsilon}) = \boldsymbol{\varepsilon}'\mathbf{M}'\mathbf{M}\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon},$$

where the last step uses $\mathbf{M}'\mathbf{M} = \mathbf{M}\mathbf{M} = \mathbf{M}$ (symmetric idempotent).

Step 2: Apply Lemma 3.6. With $\mathbf{z} = \boldsymbol{\varepsilon}$, $\mathbf{A} = \mathbf{M}$, $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ (by (A2)), and $\boldsymbol{\Sigma} = \text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ (by (A3)):

$$\mathbb{E}[\text{SSE}] = \mathbb{E}[\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}] = \text{tr}(\mathbf{M} \cdot \sigma^2 \mathbf{I}) + \mathbf{0}'\mathbf{M}\mathbf{0} = \sigma^2 \text{tr}(\mathbf{M}).$$

Step 3: Evaluate the trace. By Theorem 2.9(iv), $\text{tr}(\mathbf{M}) = n - k - 1$. Hence $\mathbb{E}[\text{SSE}] = \sigma^2(n - k - 1)$.

Step 4: Conclude.

$$\mathbb{E}[\text{MSE}] = \mathbb{E}\left[\frac{\text{SSE}}{n - k - 1}\right] = \frac{\sigma^2(n - k - 1)}{n - k - 1} = \sigma^2. \quad \square$$

3.5 Variance of Individual Residuals

Theorem 3.8 (Variance of Residuals). Under (A3) and (A4):

$$\text{var}(\mathbf{e}) = \sigma^2 \mathbf{M}, \quad \text{and in particular, } \text{var}(e_i) = \sigma^2(1 - h_{ii}). \quad (16)$$

Proof. Since $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ and $\mathbf{M}$ is symmetric ($\mathbf{M}' = \mathbf{M}$) and idempotent ($\mathbf{M}^2 = \mathbf{M}$):

$$\text{var}(\mathbf{e}) = \mathbf{M} \text{var}(\boldsymbol{\varepsilon}) \mathbf{M}' = \mathbf{M}(\sigma^2 \mathbf{I}) \mathbf{M}' = \sigma^2 \mathbf{M} \mathbf{M} = \sigma^2 \mathbf{M}^2 = \sigma^2 \mathbf{M}.$$

The (i, i) entry gives $\text{var}(e_i) = \sigma^2 m_{ii}$. Since $\mathbf{M} = \mathbf{I} - \mathbf{H}$, we have $m_{ii} = 1 - h_{ii}$, so $\text{var}(e_i) = \sigma^2(1 - h_{ii})$. $\square$

Remark 3.9. By Theorem 2.14(i), $0 \leq h_{ii} \leq 1$, so $\text{var}(e_i) \leq \sigma^2$. High-leverage observations (large h_{ii}) produce residuals with smaller variance: the fitted line is pulled toward them, leaving less room for the residual to vary.

4 Multivariate Normal Tools

This section collects the distributional results needed for exact inference in Section 5.

4.1 Prerequisites: the Multivariate Normal Distribution

For a random vector $\mathbf{z} \in \mathbb{R}^n$, the **moment-generating function** (MGF) is $M_{\mathbf{z}}(\mathbf{t}) = \mathbb{E}[\exp(\mathbf{t}'\mathbf{z})]$, $\mathbf{t} \in \mathbb{R}^n$, whenever the expectation is finite.

Definition 4.1 (Multivariate normal distribution). Let $\boldsymbol{\mu} \in \mathbb{R}^n$ and let $\boldsymbol{\Sigma}$ be a symmetric positive semi-definite $n \times n$ matrix. A random vector $\mathbf{z}$ has the **multivariate normal distribution** $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ if

$$M_{\mathbf{z}}(\mathbf{t}) = \exp(\mathbf{t}'\boldsymbol{\mu} + \frac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}\mathbf{t}) \quad \text{for all } \mathbf{t} \in \mathbb{R}^n. \quad (17)$$

Its mean vector is $\boldsymbol{\mu}$ and its covariance matrix is $\boldsymbol{\Sigma}$: the gradient of the right-hand side of (17) is $(\boldsymbol{\mu} + \boldsymbol{\Sigma}\mathbf{t}) M_{\mathbf{z}}(\mathbf{t})$ and its Hessian is $[\boldsymbol{\Sigma} + (\boldsymbol{\mu} + \boldsymbol{\Sigma}\mathbf{t})(\boldsymbol{\mu} + \boldsymbol{\Sigma}\mathbf{t})'] M_{\mathbf{z}}(\mathbf{t})$, so evaluating at $\mathbf{t} = \mathbf{0}$ and using fact (N6) below gives $\mathbb{E}[\mathbf{z}] = \boldsymbol{\mu}$ and $\mathbb{E}[\mathbf{z}\mathbf{z}'] = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}'$, whence $\text{var}(\mathbf{z}) = \mathbb{E}[\mathbf{z}\mathbf{z}'] - \boldsymbol{\mu}\boldsymbol{\mu}' = \boldsymbol{\Sigma}$. For $n = 1$ this is the usual $N(\mu, \sigma^2)$, $\sigma^2 \geq 0$.

The definition is stated through the MGF rather than through a density because $\boldsymbol{\Sigma}$ is allowed to be singular: several normal vectors below, such as the residual vector $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ with $\text{var}(\mathbf{e}) = \sigma^2 \mathbf{M}$ (Theorems 3.8 and 5.2), have singular covariance matrices and no density. The following facts about MGFs and distributions are used without proof; together with (L1)–(L4) they are the only results imported from outside these notes.

- (N1) *Uniqueness.* If two random vectors have MGFs that are finite and equal on an open neighbourhood of $\mathbf{0}$, they have the same distribution. (Used in Theorem 4.2.)
- (N2) *Independence and factorization.* Random vectors $\mathbf{U}_1, \dots, \mathbf{U}_m$ (with MGFs finite near $\mathbf{0}$) are mutually independent if and only if their joint MGF factorizes: $M_{(\mathbf{U}_1, \dots, \mathbf{U}_m)}(\mathbf{t}_1, \dots, \mathbf{t}_m) = \prod_{i=1}^m M_{\mathbf{U}_i}(\mathbf{t}_i)$ near $\mathbf{0}$. (Used in Theorems 4.3, 4.5 and 9.2.)
- (N3) *Functions of independent vectors.* If $\mathbf{U}$ and $\mathbf{V}$ are independent (written $\mathbf{U} \perp\!\!\!\perp \mathbf{V}$; the symbol $\perp$ is reserved for orthogonality of vectors and subspaces), then $f(\mathbf{U})$ and $g(\mathbf{V})$ are independent for any (measurable) functions f, g . (Used whenever “a function of $\hat{\boldsymbol{\beta}}$ is independent of SSE” is asserted, e.g. in Theorems 4.6, 5.2, 7.1, 7.4, 7.9 and 9.2.)
- (N4) *Density in the nonsingular case.* If $\boldsymbol{\Sigma}$ is positive definite, $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ has the density $f(\mathbf{z}) = (2\pi)^{-n/2} |\boldsymbol{\Sigma}|^{-1/2} \exp(-\frac{1}{2}(\mathbf{z} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{z} - \boldsymbol{\mu}))$ on $\mathbb{R}^n$. (Used in Theorem 5.3.)
- (N5) *Distributions derived from the normal.* The **chi-square distribution** χ_r^2 ($r \geq 1$) is the distribution of $\sum_{i=1}^r W_i^2$ for i.i.d. $W_i \sim N(0, 1)$. If $Z \sim N(0, 1)$ and $V \sim \chi_\nu^2$ are independent, $Z/\sqrt{V/\nu}$ has the *t distribution* t_ν . If $U \sim \chi_a^2$ and $V \sim \chi_b^2$ are independent, $(U/a)/(V/b)$ has the *F distribution* $F_{a,b}$. For $\alpha \in (0, 1)$, the **upper α quantiles** $t_{\alpha, \nu}$ and $F_{\alpha, a, b}$ are the

numbers with $P(T > t_{\alpha,\nu}) = \alpha$ for $T \sim t_\nu$ and $P(F > F_{\alpha,a,b}) = \alpha$ for $F \sim F_{a,b}$; all three have continuous distribution functions that are strictly increasing on their supports, so these quantiles exist and are unique. (Used from Theorem 4.5 onward; the t and F quantiles enter in Theorems 7.9, 7.10, 9.1 and 9.2.)

- (N6) *Moments from the MGF.* If $M_{\mathbf{z}}$ is finite on an open neighbourhood of $\mathbf{0}$, then its first- and second-order partial derivatives exist there and, at $\mathbf{t} = \mathbf{0}$, $\partial M_{\mathbf{z}} / \partial t_i = E[z_i]$ and $\partial^2 M_{\mathbf{z}} / \partial t_i \partial t_j = E[z_i z_j]$; that is, the gradient of $M_{\mathbf{z}}$ at $\mathbf{0}$ is $E[\mathbf{z}]$ and its Hessian (the matrix of second partial derivatives) is $E[\mathbf{z}\mathbf{z}']$. (Used in Definition 4.1.)
- (N7) *Independence implies uncorrelatedness.* If $\mathbf{U}$ and $\mathbf{V}$ are independent random vectors with finite second moments, then $E[\mathbf{U}\mathbf{V}'] = E[\mathbf{U}]E[\mathbf{V}]'$, i.e. $\text{cov}(\mathbf{U}, \mathbf{V}) = \mathbf{0}$ with cov as defined in Lemma 3.2. Consequently, for independent scalar random variables U and V , $\text{var}(U \pm V) = \text{var}(U) + \text{var}(V)$: apply Lemma 3.2(i) with $\mathbf{z} = (U, V)'$ and $\mathbf{A} = (1, \pm 1)$, the off-diagonal entries of $\text{var}(\mathbf{z})$ being $\text{cov}(U, V) = 0$. (Used in Theorems 4.3 and 9.2 and in the solution to Exercise B6.)

The following consequence of (N3) and (N5) is used in Theorems 7.9 and 9.2: *the t distribution is symmetric about 0.* Let $T = Z/\sqrt{V/\nu}$ with $Z \sim N(0, 1)$ and $V \sim \chi_\nu^2$ independent, as in (N5). Then $M_{-Z}(t) = E[e^{t(-Z)}] = M_Z(-t) = e^{t^2/2} = M_Z(t)$ for all t , so $-Z \sim N(0, 1)$ by Definition 4.1; $-Z$ is still independent of V ((N3)); hence $-T = (-Z)/\sqrt{V/\nu} \sim t_\nu$ by (N5), and $P(T < -c) = P(-T > c) = P(T > c)$ for every $c \in \mathbb{R}$. Taking $c = t_{\alpha,\nu}$ gives the first identity below; the second follows from the first with $\alpha/2$ in place of α , because $P(-t_{\alpha/2,\nu} \leq T \leq t_{\alpha/2,\nu}) = 1 - P(T < -t_{\alpha/2,\nu}) - P(T > t_{\alpha/2,\nu})$:

$$P(T < -t_{\alpha,\nu}) = \alpha \quad \text{and} \quad P(-t_{\alpha/2,\nu} \leq T \leq t_{\alpha/2,\nu}) = 1 - \alpha \quad (T \sim t_\nu, \alpha \in (0, 1)). \quad (18)$$

4.2 Linear Transformations, Independence, and Quadratic Forms

Theorem 4.2 (Closure Under Linear Transformations). If $\mathbf{z} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is an $n \times 1$ normal random vector, $\mathbf{A}$ is an $m \times n$ constant matrix, and $\mathbf{b} \in \mathbb{R}^m$ is a constant vector, then

$$\mathbf{Az} + \mathbf{b} \sim N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'). \quad (19)$$

Proof. By Definition 4.1, the MGF of $\mathbf{z}$ is $M_{\mathbf{z}}(\mathbf{t}) = \exp(\mathbf{t}'\boldsymbol{\mu} + \frac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}\mathbf{t})$. The MGF of $\mathbf{w} = \mathbf{Az} + \mathbf{b}$ is:

$$\begin{aligned} M_{\mathbf{w}}(\mathbf{t}) &= E\left[e^{\mathbf{t}'(\mathbf{Az} + \mathbf{b})}\right] = e^{\mathbf{t}'\mathbf{b}} \cdot E\left[e^{(\mathbf{A}'\mathbf{t})'\mathbf{z}}\right] = e^{\mathbf{t}'\mathbf{b}} \cdot M_{\mathbf{z}}(\mathbf{A}'\mathbf{t}) \\ &= e^{\mathbf{t}'\mathbf{b}} \cdot \exp\left((\mathbf{A}'\mathbf{t})'\boldsymbol{\mu} + \frac{1}{2}(\mathbf{A}'\mathbf{t})'\boldsymbol{\Sigma}(\mathbf{A}'\mathbf{t})\right) \\ &= \exp\left(\mathbf{t}'(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}) + \frac{1}{2}\mathbf{t}'\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'\mathbf{t}\right). \end{aligned}$$

This is the MGF (17) of $N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$ (note that $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'$ is symmetric positive semi-definite). By uniqueness of the MGF, fact (N1), the result follows. $\square$

Theorem 4.3 (Uncorrelatedness $\Leftrightarrow$ Independence Under Joint Normality). Let $(\mathbf{U}', \mathbf{V}')'$ be jointly normal:

$$\begin{pmatrix} \mathbf{U} \\ \mathbf{V} \end{pmatrix} \sim N\left(\begin{pmatrix} \boldsymbol{\mu}_U \\ \boldsymbol{\mu}_V \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_{UU} & \boldsymbol{\Sigma}_{UV} \\ \boldsymbol{\Sigma}_{VU} & \boldsymbol{\Sigma}_{VV} \end{pmatrix}\right).$$

Then $\boldsymbol{\Sigma}_{UV} = \mathbf{0}$ if and only if $\mathbf{U}$ and $\mathbf{V}$ are independent.

Proof. ($\Leftarrow$): Independence implies uncorrelatedness for any distribution (fact (N7)), and $\boldsymbol{\Sigma}_{UV} = \text{cov}(\mathbf{U}, \mathbf{V})$ because $\boldsymbol{\Sigma}$ is the covariance matrix of $(\mathbf{U}', \mathbf{V}')'$ (Definition 4.1); hence $\boldsymbol{\Sigma}_{UV} = \mathbf{0}$.

($\Rightarrow$): With $\boldsymbol{\Sigma}_{UV} = \mathbf{0}$, the covariance is block-diagonal. The joint MGF is

$$M_{(\mathbf{U}, \mathbf{V})}(\mathbf{s}, \mathbf{t}) = \exp\left((\mathbf{s}', \mathbf{t}')\boldsymbol{\mu} + \frac{1}{2}(\mathbf{s}', \mathbf{t}')\boldsymbol{\Sigma}(\mathbf{s}', \mathbf{t}')'\right).$$

Block expansion gives

$$(\mathbf{s}', \mathbf{t}')\boldsymbol{\mu} = \mathbf{s}'\boldsymbol{\mu}_U + \mathbf{t}'\boldsymbol{\mu}_V, \quad (\mathbf{s}', \mathbf{t}')\boldsymbol{\Sigma} \begin{pmatrix} \mathbf{s} \\ \mathbf{t} \end{pmatrix} = \mathbf{s}'\boldsymbol{\Sigma}_{UU}\mathbf{s} + \mathbf{t}'\boldsymbol{\Sigma}_{VV}\mathbf{t},$$

using $\boldsymbol{\Sigma}_{UV} = \boldsymbol{\Sigma}'_{VU} = \mathbf{0}$. The joint MGF factors:

$$M_{(\mathbf{U}, \mathbf{V})}(\mathbf{s}, \mathbf{t}) = \exp(\mathbf{s}'\boldsymbol{\mu}_U + \frac{1}{2}\mathbf{s}'\boldsymbol{\Sigma}_{UU}\mathbf{s}) \cdot \exp(\mathbf{t}'\boldsymbol{\mu}_V + \frac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}_{VV}\mathbf{t}) = M_{\mathbf{U}}(\mathbf{s})M_{\mathbf{V}}(\mathbf{t}),$$

where the last step identifies the two factors as the marginal MGFs by setting $\mathbf{t} = \mathbf{0}$, respectively $\mathbf{s} = \mathbf{0}$, in the joint MGF. By fact (N2), this factorization implies independence. $\square$

Remark 4.4. For general (non-normal) distributions, uncorrelatedness does *not* imply independence. The equivalence is a special property of the multivariate normal.

Theorem 4.5 (Chi-Square Distribution of Quadratic Forms). Let $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and $\mathbf{A}$ be an $n \times n$ symmetric idempotent matrix with $\mathbf{A} \neq \mathbf{0}$ (so that $r = \text{rank}(\mathbf{A}) \geq 1$, as the definition of χ_r^2 in (N5) requires). Then

$$\mathbf{z}'\mathbf{A}\mathbf{z} \sim \chi_r^2, \quad \text{where } r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}). \quad (20)$$

Proof. Step 1: Spectral decomposition. Since $\mathbf{A}$ is symmetric, there exists an orthogonal matrix $\mathbf{P}$ ($\mathbf{P}'\mathbf{P} = \mathbf{P}\mathbf{P}' = \mathbf{I}$) such that $\mathbf{A} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}'$, where $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$. By idempotency, each $\lambda_i \in \{0, 1\}$ (Step 1 of the proof of Lemma 2.8). Without loss of generality, let the first r eigenvalues be 1 and the rest be 0:

$$\boldsymbol{\Lambda} = \text{diag}(\underbrace{1, \dots, 1}_r, \underbrace{0, \dots, 0}_{n-r}).$$

Step 2: Apply orthogonal transformation. Let $\mathbf{w} = \mathbf{P}'\mathbf{z}$. Since $\mathbf{P}$ is orthogonal:

$$\mathbb{E}[\mathbf{w}] = \mathbf{P}'\mathbb{E}[\mathbf{z}] = \mathbf{0}, \quad \text{var}(\mathbf{w}) = \mathbf{P}'\mathbf{I}\mathbf{P} = \mathbf{I}_n.$$

By Theorem 4.2, $\mathbf{w} \sim N(\mathbf{0}, \mathbf{I}_n)$. Its MGF $\exp(\frac{1}{2}\mathbf{t}'\mathbf{t}) = \prod_{i=1}^n \exp(\frac{1}{2}t_i^2)$ is the product of n standard normal MGFs, so by fact (N2) (the n -block form of the argument in Theorem 4.3) $w_1, \dots, w_n$ are i.i.d. $N(0, 1)$.

Step 3: Evaluate the quadratic form.

$$\mathbf{z}'\mathbf{A}\mathbf{z} = \mathbf{z}'\mathbf{P}\boldsymbol{\Lambda}\mathbf{P}'\mathbf{z} = (\mathbf{P}'\mathbf{z})'\boldsymbol{\Lambda}(\mathbf{P}'\mathbf{z}) = \mathbf{w}'\boldsymbol{\Lambda}\mathbf{w} = \sum_{i=1}^n \lambda_i w_i^2 = \sum_{i=1}^r w_i^2.$$

Step 4: Conclude. Since $w_1, \dots, w_r$ are independent $N(0, 1)$, by definition $\sum_{i=1}^r w_i^2 \sim \chi_r^2$. $\square$

Theorem 4.6 (Independence of Quadratic Forms — a Special Case of Craig's Theorem). Let $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$, and let $\mathbf{A}_1, \mathbf{A}_2$ be $n \times n$ symmetric idempotent matrices with $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$. Then $\mathbf{z}'\mathbf{A}_1\mathbf{z}$ and $\mathbf{z}'\mathbf{A}_2\mathbf{z}$ are independent.

Proof. Define $\mathbf{U} = \mathbf{A}_1\mathbf{z}$ and $\mathbf{V} = \mathbf{A}_2\mathbf{z}$. Stack them into a single vector obtained from $\mathbf{z}$ by one linear map:

$$\begin{pmatrix} \mathbf{U} \\ \mathbf{V} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{pmatrix} \mathbf{z}.$$

Since $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$, Theorem 4.2 implies $(\mathbf{U}', \mathbf{V}')'$ is *jointly* normal. Their cross-covariance is, by Lemma 3.2(ii),

$$\text{cov}(\mathbf{U}, \mathbf{V}) = \mathbf{A}_1 \text{var}(\mathbf{z}) \mathbf{A}_2' = \mathbf{A}_1 \mathbf{I} \mathbf{A}_2 = \mathbf{A}_1 \mathbf{A}_2 = \mathbf{0},$$

where $\mathbf{A}_2' = \mathbf{A}_2$ by symmetry. By Theorem 4.3, $\mathbf{U}$ and $\mathbf{V}$ are independent. By symmetric idempotency of $\mathbf{A}_1$, $\mathbf{z}'\mathbf{A}_1\mathbf{z} = \mathbf{z}'\mathbf{A}_1'\mathbf{A}_1\mathbf{z} = \|\mathbf{U}\|^2$ is a function of $\mathbf{U}$ alone; similarly $\mathbf{z}'\mathbf{A}_2\mathbf{z} = \|\mathbf{V}\|^2$ depends only on $\mathbf{V}$. The two quadratic forms are therefore independent (fact (N3)). $\square$

Remark 4.7. In Lemma 7.2 (Section 7), Theorem 4.6 is applied with $\mathbf{A}_1 = \mathbf{H} - n^{-1}\mathbf{1}\mathbf{1}'$ and $\mathbf{A}_2 = \mathbf{M}$ to prove $\text{SSR} \perp\!\!\!\perp \text{SSE}$.

5 Exact Inference Under Normality

In this section and in Sections 7 and 9, all five assumptions (A1)–(A5) are in force, with two exceptions noted where they occur: the identities of Corollaries 7.6 and 7.8 are algebraic and, like Theorem 10.3, use only (A4) and $\text{rank}(\mathbf{R}) = q$. Sections 6, 8 and 10, by contrast, consist of algebraic identities that hold for every realization of $\mathbf{Y}$ and use only the standing assumption (A4) (Theorem 8.3 also uses (A2)); this is restated at the start of each of those sections.

5.1 Sampling Distribution of $\hat{\beta}$

Theorem 5.1 (Normality of $\hat{\beta}$). Under (A1)–(A5):

$$\boxed{\hat{\beta} \sim N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})}. \quad (21)$$

Proof. From (12), $\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon$. This is an affine transformation of ϵ . Under (A5), $\epsilon \sim N(\mathbf{0}, \sigma^2\mathbf{I})$. By Theorem 4.2 with $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, $\mathbf{z} = \epsilon$, $\mathbf{b} = \beta$:

$$\begin{aligned} \hat{\beta} &\sim N(\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \cdot \mathbf{0}, (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}) \\ &= N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}). \end{aligned} \quad \square$$

5.2 Distribution of SSE/σ^2 and Independence from $\hat{\beta}$

Theorem 5.2 (Chi-Square Distribution of SSE/σ^2). Under (A1)–(A5):

$$\frac{\text{SSE}}{\sigma^2} \sim \chi_{n-k-1}^2, \quad (22)$$

and SSE is independent of $\hat{\beta}$.

Proof. Step 1: Standardize. Let $\mathbf{z} = \epsilon/\sigma$. Under (A5), $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$. Since $\mathbf{e} = \mathbf{M}\epsilon$:

$$\frac{\text{SSE}}{\sigma^2} = \frac{\epsilon'\mathbf{M}\epsilon}{\sigma^2} = \mathbf{z}'\mathbf{M}\mathbf{z}.$$

Step 2: Apply Theorem 4.5. The matrix $\mathbf{M}$ is symmetric and idempotent (Theorem 2.9(i)), with $\text{tr}(\mathbf{M}) = n - k - 1$ (Theorem 2.9(iv)). Hence $\mathbf{z}'\mathbf{M}\mathbf{z} \sim \chi_{n-k-1}^2$.

Step 3: Prove independence. Stack $\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon$ and $\mathbf{e} = \mathbf{M}\epsilon$ into a single vector obtained from ϵ by one linear map:

$$\begin{pmatrix} \hat{\beta} - \beta \\ \mathbf{e} \end{pmatrix} = \begin{pmatrix} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ \mathbf{M} \end{pmatrix} \epsilon.$$

Since $\epsilon \sim N(\mathbf{0}, \sigma^2\mathbf{I})$, Theorem 4.2 makes this stacked vector *jointly* normal (individual normality alone would not suffice; joint normality requires that both pieces come from a single normal vector). Their cross-covariance is, by Lemma 3.2(ii),

$$\begin{aligned} \text{cov}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon, \mathbf{M}\epsilon) &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \text{var}(\epsilon)\mathbf{M}' \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{M} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} = \mathbf{0}, \end{aligned}$$

using $\mathbf{X}'\mathbf{M} = (\mathbf{M}\mathbf{X})' = \mathbf{0}$ (Theorem 2.9(ii)). By Theorem 4.3, $\hat{\beta} - \beta$ and $\mathbf{e}$ are independent. Since $\hat{\beta}$ is $(\hat{\beta} - \beta)$ plus a constant and $\text{SSE} = \mathbf{e}'\mathbf{e}$ is a function of $\mathbf{e}$, $\hat{\beta}$ is independent of $\mathbf{e}$ (hence of SSE). $\square$

5.3 OLS as Maximum Likelihood Under Normality

Theorem 5.3 (OLS Equals MLE Under Normality). Under (A1)–(A5), the OLS estimator $\hat{\beta}$ is also the maximum likelihood estimator of β . The MLE of σ^2 is $\hat{\sigma}_{\text{MLE}}^2 = \text{SSE}/n$, which is biased.

Proof. Under (A5), $\varepsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$, so by Theorem 4.2 applied to the affine transformation $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$, $\mathbf{Y} \sim N(\mathbf{X}\beta, \sigma^2 \mathbf{I})$. Since $\sigma^2 \mathbf{I}$ is positive definite, $\mathbf{Y}$ has the density of fact (N4) with $|\sigma^2 \mathbf{I}| = (\sigma^2)^n$, and the log-likelihood is:

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta).$$

Maximizing over β : In $\ell(\beta, \sigma^2)$, the first two terms do not involve β , and the last term is $-(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta)/(2\sigma^2)$ with $\sigma^2 > 0$ treated as fixed. Since $-1/(2\sigma^2) < 0$, maximizing ℓ over β is equivalent to minimizing $(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) = Q(\beta)$. This is exactly the OLS problem, so $\hat{\beta}_{\text{MLE}} = \hat{\beta}_{\text{OLS}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$.

Maximizing over σ^2 : Substituting $\hat{\beta}$ and noting that $(\mathbf{Y} - \mathbf{X}\hat{\beta})'(\mathbf{Y} - \mathbf{X}\hat{\beta}) = \text{SSE}$, the profile log-likelihood becomes

$$\ell(\hat{\beta}, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{\text{SSE}}{2\sigma^2}.$$

Differentiating with respect to σ^2 :

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\text{SSE}}{2(\sigma^2)^2} = 0 \implies \hat{\sigma}_{\text{MLE}}^2 = \frac{\text{SSE}}{n}.$$

Second-order condition: Writing $u = \sigma^2$ for brevity,

$$\frac{\partial^2 \ell}{\partial u^2} = \frac{n}{2u^2} - \frac{\text{SSE}}{u^3}.$$

At the critical point $\hat{u} = \text{SSE}/n$, so $u^2 = \text{SSE}^2/n^2$ and $u^3 = \text{SSE}^3/n^3$:

$$\left. \frac{\partial^2 \ell}{\partial u^2} \right|_{\hat{u}} = \frac{n \cdot n^2}{2\text{SSE}^2} - \frac{\text{SSE} \cdot n^3}{\text{SSE}^3} = \frac{n^3}{2\text{SSE}^2} - \frac{n^3}{\text{SSE}^2} = -\frac{n^3}{2\text{SSE}^2} < 0,$$

confirming a local maximum. It is the global maximum over $u \in (0, \infty)$: here $\text{SSE} > 0$ (this holds with probability one, since $\text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ by Theorem 5.2 is a continuous distribution; if $\text{SSE} = 0$ the likelihood is unbounded and no MLE of σ^2 exists), so $\ell(\hat{\beta}, u) \rightarrow -\infty$ both as $u \rightarrow 0^+$ (the term $-\text{SSE}/(2u)$ dominates) and as $u \rightarrow \infty$ (the term $-\frac{n}{2} \ln u$ dominates). A continuous function on $(0, \infty)$ that tends to $-\infty$ at both ends attains its supremum at a critical point, and $\hat{u}$ is the only one. Finally, since for every fixed u the maximizer over β is $\hat{\beta}$ (the unique global minimizer of Q , Theorem 2.4), $(\hat{\beta}, \hat{u})$ maximizes ℓ jointly.

Bias of $\hat{\sigma}_{\text{MLE}}^2$. Using Theorem 3.7 ($\mathbb{E}[\text{SSE}] = (n - k - 1)\sigma^2$):

$$\mathbb{E}[\hat{\sigma}_{\text{MLE}}^2] = \mathbb{E}\left[\frac{\text{SSE}}{n}\right] = \frac{(n - k - 1)\sigma^2}{n} = \frac{n - k - 1}{n}\sigma^2 < \sigma^2,$$

so $\hat{\sigma}_{\text{MLE}}^2$ underestimates σ^2 on average by a factor of $(n - k - 1)/n$. The unbiased estimator is $\text{MSE} = \text{SSE}/(n - k - 1)$, which rescales by this factor. $\square$

6 Sum of Squares and the Coefficient of Determination

The results of this section are algebraic identities about the fitted values and residuals of a given data set. They use only the standing assumption (A4) (applied to every design matrix involved) and, where stated, the presence of an intercept column; none of (A1)–(A3) or (A5) is used. (Theorem 6.10 refers to the t statistic of Section 7 only as a name for an algebraic ratio.)

6.1 Variance Decomposition

Definition 6.1 (Centering Matrix). Let $\mathbf{J} = \frac{1}{n}\mathbf{1}\mathbf{1}' = \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}'$, the hat matrix (Definition 2.6) of the intercept-only design $\mathbf{X} = \mathbf{1}$; by Remark 2.10 it is the orthogonal projection onto $\mathcal{C}(\mathbf{1})$, and $\mathbf{J}\mathbf{v} = \bar{v}\mathbf{1}$ for any vector $\mathbf{v}$, where $\bar{v} = \frac{1}{n}\mathbf{1}'\mathbf{v}$. The **centering matrix** is the corresponding residual maker $\mathbf{M}_0 = \mathbf{I}_n - \mathbf{J} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}'$. For any vector $\mathbf{v}$, $\mathbf{M}_0\mathbf{v} = \mathbf{v} - \bar{v}\mathbf{1}$.

Remark 6.2 (Notation: $\mathbf{M}_0$ versus $\mathbf{M}$). The subscript “0” in $\mathbf{M}_0$ signals that it is the residual maker with respect to the *intercept only* (the single column $\mathbf{1}$), whereas $\mathbf{M} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ from Definition 2.6 is the residual maker with respect to the *full* design matrix $\mathbf{X}$. Conceptually,

$\mathbf{M}_0$ centers a vector, $\mathbf{M}$ regresses out all of $\mathbf{X}$.

Both are symmetric idempotent, but they project onto different subspaces:

$$\mathbf{M}_0 : \mathbb{R}^n \rightarrow \mathbf{1}^\perp, \quad \mathbf{M} : \mathbb{R}^n \rightarrow \mathcal{C}(\mathbf{X})^\perp.$$

If the model includes an intercept then $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, hence $\mathcal{C}(\mathbf{X})^\perp \subseteq \mathbf{1}^\perp$, and so $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0\mathbf{M} = \mathbf{M}$ (this follows from the projection identity $\mathbf{H}\mathbf{J} = \mathbf{J}\mathbf{H} = \mathbf{J}$ proved in Lemma 7.2, Step 2, by taking complements). In particular, $\mathbf{M}$ and $\mathbf{M}_0$ are not equal, nor is one a simple scalar multiple of the other; they coincide only in the trivial case $\mathbf{X} = \mathbf{1}$.

Remark 6.3. $\mathbf{M}_0$ is symmetric and idempotent with $\text{tr}(\mathbf{M}_0) = n - 1$: it is the residual maker $\mathbf{M}$ of the intercept-only design $\mathbf{X} = \mathbf{1}$, so Theorem 2.9(i) and (iv) apply with $k = 0$.

Theorem 6.4 (Variance Decomposition). In the model with an intercept, the sums of squares introduced in Definition 3.1 satisfy:

$$\boxed{\text{SST} = \text{SSR} + \text{SSE}}, \tag{23}$$

with matrix expressions:

- $\text{SST} = \sum (Y_i - \bar{Y})^2 = \mathbf{Y}'\mathbf{M}_0\mathbf{Y}$ (Total Sum of Squares),
- $\text{SSR} = \sum (\hat{Y}_i - \bar{Y})^2 = \hat{\mathbf{Y}}'\mathbf{M}_0\hat{\mathbf{Y}}$ (Regression Sum of Squares),
- $\text{SSE} = \sum (Y_i - \hat{Y}_i)^2 = \mathbf{e}'\mathbf{e}$ (Error/Residual Sum of Squares).

Proof. Step 1: Verify the matrix expressions for SST and SSR. For any vector $\mathbf{v} \in \mathbb{R}^n$, Definition 6.1 gives $\mathbf{M}_0\mathbf{v} = \mathbf{v} - \bar{v}\mathbf{1}$, so

$$\mathbf{v}'\mathbf{M}_0\mathbf{v} = (\mathbf{M}_0\mathbf{v})'(\mathbf{M}_0\mathbf{v}) = \|\mathbf{v} - \bar{v}\mathbf{1}\|^2 = \sum_{i=1}^n (v_i - \bar{v})^2,$$

using $\mathbf{M}_0'\mathbf{M}_0 = \mathbf{M}_0^2 = \mathbf{M}_0$ (symmetric idempotent). Applying this with $\mathbf{v} = \mathbf{Y}$ gives $\text{SST} = \mathbf{Y}'\mathbf{M}_0\mathbf{Y}$; with $\mathbf{v} = \hat{\mathbf{Y}}$ gives $\text{SSR} = \hat{\mathbf{Y}}'\mathbf{M}_0\hat{\mathbf{Y}}$, provided $\bar{\hat{Y}} = \bar{Y}$ —which follows from $\mathbf{1}'\mathbf{e} = 0$ (Theorem 2.12(ii)) since $\mathbf{1}'\hat{\mathbf{Y}} = \mathbf{1}'(\mathbf{Y} - \mathbf{e}) = \mathbf{1}'\mathbf{Y}$.

Step 2: Decompose $\mathbf{M}_0\mathbf{Y}$ into orthogonal components. Write

$$\mathbf{M}_0\mathbf{Y} = \mathbf{M}_0(\hat{\mathbf{Y}} + \mathbf{e}) = \mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{M}_0\mathbf{e} = \mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{e}, \tag{24}$$

where the last equality uses $\mathbf{M}_0\mathbf{e} = \mathbf{e} - \bar{e}\mathbf{1} = \mathbf{e}$ because $\bar{e} = n^{-1}\mathbf{1}'\mathbf{e} = 0$ (Theorem 2.12(ii)).

Step 3: Expand SST. Using Step 2 and $\mathbf{M}_0^2 = \mathbf{M}_0$:

$$\begin{aligned} \text{SST} &= \mathbf{Y}'\mathbf{M}_0\mathbf{Y} = (\mathbf{M}_0\mathbf{Y})'(\mathbf{M}_0\mathbf{Y}) = (\mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{e})'(\mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{e}) \\ &= \text{SSR} + 2\mathbf{e}'\mathbf{M}_0\hat{\mathbf{Y}} + \text{SSE}. \end{aligned}$$

Step 4: The cross term vanishes. Using $\mathbf{e}'\hat{\mathbf{Y}} = 0$ and $\mathbf{e}'\mathbf{1} = 0$ (Theorem 2.12(ii), (iii)):

$$\mathbf{e}'\mathbf{M}_0\hat{\mathbf{Y}} = \mathbf{e}'\hat{\mathbf{Y}} - \bar{e}\mathbf{e}'\mathbf{1} = 0.$$

Hence $\text{SST} = \text{SSR} + \text{SSE}$. □

Remark 6.5 (Geometric view). Equation (24) is an orthogonal decomposition of $\mathbf{M}_0\mathbf{Y}$ into two pieces: the centered fitted values $\mathbf{M}_0\hat{\mathbf{Y}}$, which lie in the subspace $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$ (the non-intercept directions of the column space), and the residuals $\mathbf{e}$, which lie in $\mathcal{C}(\mathbf{X})^\perp$. Pythagoras applied to these two orthogonal pieces is precisely $\text{SST} = \text{SSR} + \text{SSE}$.

6.2 Coefficient of Determination

Definition 6.6 (R^2 and Adjusted R^2). Assume the model has an intercept and $\text{SST} > 0$, i.e. $\mathbf{Y}$ is not a constant vector ($\text{SST} = \|\mathbf{M}_0\mathbf{Y}\|^2 = 0$ iff $\mathbf{Y} = \bar{Y}\mathbf{1}$). The **coefficient of determination** is

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}. \quad (25)$$

It measures the proportion of total variation in $\mathbf{Y}$ explained by the regression. The **adjusted** R^2 is

$$\bar{R}^2 = 1 - \frac{\text{SSE}/(n - k - 1)}{\text{SST}/(n - 1)} = 1 - \frac{n - 1}{n - k - 1}(1 - R^2) = 1 - \frac{\text{MSE}}{s_Y^2}, \quad (26)$$

where $s_Y^2 = \text{SST}/(n - 1)$. The adjusted R^2 can be negative.

Proposition 6.7 (Bounds on R^2). In any model with an intercept and $\text{SST} > 0$, $0 \leq R^2 \leq 1$.

Proof. $R^2 = \text{SSR}/\text{SST} \geq 0$ because $\text{SSR} = \sum(\hat{Y}_i - \bar{Y})^2 \geq 0$ and $\text{SST} = \sum(Y_i - \bar{Y})^2 > 0$ by hypothesis. For the upper bound, Theorem 6.4 gives $\text{SST} = \text{SSR} + \text{SSE}$ with $\text{SSE} \geq 0$, so $\text{SSR} \leq \text{SST}$, i.e., $R^2 \leq 1$. $\square$

Theorem 6.8 (Monotonicity of R^2). Adding a regressor to the model cannot decrease R^2 . That is, if R_k^2 is the coefficient of determination with k regressors and R_{k+1}^2 is that with an additional regressor, then $R_{k+1}^2 \geq R_k^2$.

Proof. Let $\mathcal{C}_k \subseteq \mathcal{C}_{k+1}$ denote the column spaces of the smaller and larger design matrices (the inclusion holds because every column of $\mathbf{X}_k$ is also a column of $\mathbf{X}_{k+1}$). Write $\hat{\mathbf{Y}}_k, \hat{\mathbf{Y}}_{k+1}$ for the corresponding OLS fits.

By Theorem 2.5, $\hat{\mathbf{Y}}_{k+1}$ is the point in $\mathcal{C}_{k+1}$ closest to $\mathbf{Y}$, so $\|\mathbf{Y} - \hat{\mathbf{Y}}_{k+1}\| \leq \|\mathbf{Y} - \mathbf{v}\|$ for every $\mathbf{v} \in \mathcal{C}_{k+1}$. In particular, $\hat{\mathbf{Y}}_k \in \mathcal{C}_k \subseteq \mathcal{C}_{k+1}$, so

$$\text{SSE}_{k+1} = \|\mathbf{Y} - \hat{\mathbf{Y}}_{k+1}\|^2 \leq \|\mathbf{Y} - \hat{\mathbf{Y}}_k\|^2 = \text{SSE}_k.$$

Since SST does not depend on the regressors,

$$R_{k+1}^2 = 1 - \frac{\text{SSE}_{k+1}}{\text{SST}} \geq 1 - \frac{\text{SSE}_k}{\text{SST}} = R_k^2. \quad \square$$

Lemma 6.9 (Reduction in SSE from adding one regressor). Write the enlarged design as $\mathbf{X}_{k+1} = [\mathbf{X}_k \ \mathbf{x}]$, where $\mathbf{x}$ is the new regressor column, and assume $\mathbf{x}'\mathbf{M}_k\mathbf{x} > 0$ (where $\mathbf{M}_k$ is the residual maker for $\mathbf{X}_k$), which holds whenever $\mathbf{x}$ is not in $\mathcal{C}(\mathbf{X}_k)$. Then the OLS coefficient on $\mathbf{x}$ in the enlarged model is

$$\hat{\beta}_{k+1} = \frac{\mathbf{x}'\mathbf{M}_k\mathbf{Y}}{\mathbf{x}'\mathbf{M}_k\mathbf{x}}, \quad (27)$$

and the reduction in residual sum of squares is

$$\text{SSE}_k - \text{SSE}_{k+1} = \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x} = \frac{(\mathbf{x}'\mathbf{M}_k\mathbf{Y})^2}{\mathbf{x}'\mathbf{M}_k\mathbf{x}} \geq 0. \quad (28)$$

This identity uses only normal equations (no forward dependence on Section 8).

Proof. The partitioned normal equations $\mathbf{X}'_{k+1}\mathbf{X}_{k+1}\hat{\boldsymbol{\beta}}_{k+1} = \mathbf{X}'_{k+1}\mathbf{Y}$ read

$$\mathbf{X}'_k\mathbf{X}_k\hat{\boldsymbol{\beta}}_k^* + \mathbf{X}'_k\mathbf{x}\hat{\beta}_{k+1} = \mathbf{X}'_k\mathbf{Y}, \quad (\text{a})$$

$$\mathbf{x}'\mathbf{X}_k\hat{\boldsymbol{\beta}}_k^* + \mathbf{x}'\mathbf{x}\hat{\beta}_{k+1} = \mathbf{x}'\mathbf{Y}, \quad (\text{b})$$

where $\hat{\boldsymbol{\beta}}_k^*$ denotes the $\mathbf{X}_k$ -block of the enlarged estimator (its first $k+1$ coordinates, the coefficients on the columns of $\mathbf{X}_k$; generally different from the short-regression estimator $\hat{\boldsymbol{\beta}}_k$). From (a), $\hat{\boldsymbol{\beta}}_k^* = (\mathbf{X}'_k\mathbf{X}_k)^{-1}\mathbf{X}'_k(\mathbf{Y} - \mathbf{x}\hat{\beta}_{k+1})$. Substituting into (b):

$$\mathbf{x}'\mathbf{X}_k(\mathbf{X}'_k\mathbf{X}_k)^{-1}\mathbf{X}'_k(\mathbf{Y} - \mathbf{x}\hat{\beta}_{k+1}) + \mathbf{x}'\mathbf{x}\hat{\beta}_{k+1} = \mathbf{x}'\mathbf{Y}.$$

Writing $\mathbf{H}_k = \mathbf{X}_k(\mathbf{X}'_k\mathbf{X}_k)^{-1}\mathbf{X}'_k = \mathbf{I} - \mathbf{M}_k$ and rearranging:

$$\mathbf{x}'\mathbf{H}_k\mathbf{Y} - \mathbf{x}'\mathbf{H}_k\mathbf{x}\hat{\beta}_{k+1} + \mathbf{x}'\mathbf{x}\hat{\beta}_{k+1} = \mathbf{x}'\mathbf{Y} \iff \mathbf{x}'(\mathbf{I} - \mathbf{H}_k)\mathbf{x}\hat{\beta}_{k+1} = \mathbf{x}'(\mathbf{I} - \mathbf{H}_k)\mathbf{Y},$$

which gives (27).

For (28), relate $\hat{\boldsymbol{\beta}}_k^*$ to the short-regression estimator $\hat{\boldsymbol{\beta}}_k = (\mathbf{X}'_k\mathbf{X}_k)^{-1}\mathbf{X}'_k\mathbf{Y}$: from (a),

$$\hat{\boldsymbol{\beta}}_k^* = \hat{\boldsymbol{\beta}}_k - (\mathbf{X}'_k\mathbf{X}_k)^{-1}\mathbf{X}'_k\mathbf{x}\hat{\beta}_{k+1}.$$

Multiplying by $\mathbf{X}_k$ gives $\mathbf{X}_k\hat{\boldsymbol{\beta}}_k^* = \hat{\mathbf{Y}}_k - \mathbf{H}_k\mathbf{x}\hat{\beta}_{k+1}$, so

$$\hat{\mathbf{Y}}_{k+1} = \mathbf{X}_k\hat{\boldsymbol{\beta}}_k^* + \mathbf{x}\hat{\beta}_{k+1} = \hat{\mathbf{Y}}_k + (\mathbf{I} - \mathbf{H}_k)\mathbf{x}\hat{\beta}_{k+1} = \hat{\mathbf{Y}}_k + \mathbf{M}_k\mathbf{x}\hat{\beta}_{k+1},$$

and the residuals satisfy $\mathbf{e}_{k+1} = \mathbf{e}_k - \mathbf{M}_k\mathbf{x}\hat{\beta}_{k+1}$. Taking squared norms, using $\mathbf{e}_k = \mathbf{M}_k\mathbf{Y}$ and $\mathbf{M}_k^2 = \mathbf{M}_k$:

$$\text{SSE}_{k+1} = \text{SSE}_k - 2\hat{\beta}_{k+1} \cdot \mathbf{Y}'\mathbf{M}_k\mathbf{x} + \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}.$$

Substituting $\hat{\beta}_{k+1} = (\mathbf{x}'\mathbf{M}_k\mathbf{Y})/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$ makes the middle term $-2\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}$, so

$$\text{SSE}_{k+1} = \text{SSE}_k - \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x},$$

which is (28). The non-negativity reflects Theorem 6.8. $\square$

Theorem 6.10 ($\bar{R}^2$ Increases iff $|t| > 1$). Let $\mathbf{X}_{k+1} = [\mathbf{X}_k \ \mathbf{x}]$ as in Lemma 6.9, with $\mathbf{x} \notin \mathcal{C}(\mathbf{X}_k)$ (so that $\mathbf{x}'\mathbf{M}_k\mathbf{x} > 0$ and $\mathbf{X}_{k+1}$ has full column rank), and assume $n > k+2$, $\text{SST} > 0$ and $\text{SSE}_{k+1} > 0$, so that both adjusted R^2 's and the t statistic below are defined. Then adding the regressor $\mathbf{x}$ increases $\bar{R}^2$ if and only if its t -statistic in the enlarged model exceeds 1 in absolute value.

Proof. Let the model with k regressors have $\text{MSE}_k = \text{SSE}_k/(n-k-1)$, and the model with an additional regressor have $\text{MSE}_{k+1} = \text{SSE}_{k+1}/(n-k-2)$. Since $\bar{R}^2 = 1 - \text{MSE}/s_Y^2$ with the same $s_Y^2 = \text{SST}/(n-1) > 0$ in both models, $\bar{R}_{k+1}^2 > \bar{R}_k^2$ if and only if $\text{MSE}_{k+1} < \text{MSE}_k$, i.e.,

$$\frac{\text{SSE}_{k+1}}{n-k-2} < \frac{\text{SSE}_k}{n-k-1}. \quad (29)$$

Step 1: Apply Lemma 6.9. By (28), $\text{SSE}_k = \text{SSE}_{k+1} + \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}$. Substitute this into (29):

$$\begin{aligned} \frac{\text{SSE}_{k+1}}{n-k-2} &< \frac{\text{SSE}_{k+1} + \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}}{n-k-1} \\ (n-k-1)\text{SSE}_{k+1} &< (n-k-2)[\text{SSE}_{k+1} + \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}] \\ \text{SSE}_{k+1} &< (n-k-2) \cdot \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}. \end{aligned}$$

Dividing by $\text{SSE}_{k+1} = (n - k - 2)\text{MSE}_{k+1} > 0$:

$$1 < \frac{\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}}{\text{MSE}_{k+1}}. \quad (30)$$

Step 2: Identify the right-hand side as t^2 . The t -statistic for $\hat{\beta}_{k+1}$ in the enlarged model is $t_{k+1} = \hat{\beta}_{k+1}/\text{SE}(\hat{\beta}_{k+1})$ with $\text{SE}(\hat{\beta}_{k+1})^2 = \text{MSE}_{k+1} \cdot [(\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}]_{k+1,k+1}$ (defined formally in Theorem 7.1; here $[\cdot]_{k+1,k+1}$ is the diagonal entry belonging to β_{k+1} , the last one). We claim

$$[(\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}]_{k+1,k+1} = (\mathbf{x}'\mathbf{M}_k\mathbf{x})^{-1}. \quad (31)$$

Indeed, let $\mathbf{G} = (\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}\mathbf{X}'_{k+1}$, so that $\hat{\beta}_{k+1} = \mathbf{G}\mathbf{Y}$ for every $\mathbf{Y} \in \mathbb{R}^n$. Lemma 6.9 says that the last coordinate of $\mathbf{G}\mathbf{Y}$ is $\mathbf{x}'\mathbf{M}_k\mathbf{Y}/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$ for every $\mathbf{Y}$, so the last row of $\mathbf{G}$ is $\mathbf{g}' := \mathbf{x}'\mathbf{M}_k/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$. On the other hand $\mathbf{G}\mathbf{G}' = (\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}\mathbf{X}'_{k+1}\mathbf{X}_{k+1}(\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1} = (\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}$, whose last diagonal entry is the squared norm of the last row of $\mathbf{G}$:

$$[(\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}]_{k+1,k+1} = \mathbf{g}'\mathbf{g} = \frac{\mathbf{x}'\mathbf{M}_k\mathbf{M}_k\mathbf{x}}{(\mathbf{x}'\mathbf{M}_k\mathbf{x})^2} = \frac{1}{\mathbf{x}'\mathbf{M}_k\mathbf{x}},$$

using $\mathbf{M}'_k = \mathbf{M}_k = \mathbf{M}_k^2$. (Equivalently: $\text{var}(\hat{\beta}_{k+1})$ computed from (27) via Lemma 3.2 is $\sigma^2/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$, and computed from Theorem 3.4 it is $\sigma^2[(\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}]_{k+1,k+1}$; this is the block-inverse, or Schur complement, formula for the last diagonal entry, obtained here from Lemma 6.9 alone.) Hence $\text{SE}(\hat{\beta}_{k+1})^2 = \text{MSE}_{k+1}/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$, so

$$t_{k+1}^2 = \frac{\hat{\beta}_{k+1}^2}{\text{SE}(\hat{\beta}_{k+1})^2} = \frac{\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}}{\text{MSE}_{k+1}}.$$

Substituting into (30) gives $t_{k+1}^2 > 1$, i.e., $|t_{k+1}| > 1$. □

Remark 6.11 (Logical status of the forward reference). The proof refers to one later statement: the form of $\text{SE}(\hat{\beta}_{k+1})$ in Theorem 7.1, which is a *definition* of the t statistic rather than a result whose proof is needed here. It is used only to *interpret* the ratio $\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}/\text{MSE}_{k+1}$ as t_{k+1}^2 ; equation (30) and the identity (31) are self-contained algebraic consequences of Lemma 6.9. A reader seeking strict linearity can defer Theorem 6.10 until after Section 7.

7 Hypothesis Testing and Confidence Regions

7.1 Individual Coefficient t Test

Theorem 7.1 (Individual t Test). Under (A1)–(A5), for testing $H_0: \beta_j = \beta_j^0$:

$$t = \frac{\hat{\beta}_j - \beta_j^0}{\text{SE}(\hat{\beta}_j)} \sim t_{n-k-1} \quad \text{under } H_0, \quad (32)$$

where $\text{SE}(\hat{\beta}_j) = s\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$ and $s = \sqrt{\text{MSE}}$.

Proof. Step 1: Marginal distribution of $\hat{\beta}_j$. By Theorem 5.1, $\hat{\beta} \sim N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$. The j -th coordinate of a multivariate normal vector is itself normal, with mean β_j and variance equal to the (j, j) -entry of the covariance matrix (this is Theorem 4.2 applied with $\mathbf{A} = \boldsymbol{\iota}'_j$, the transpose of the j -th standard basis vector):

$$\hat{\beta}_j = \boldsymbol{\iota}'_j\hat{\beta} \sim N(\beta_j, \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}).$$

Standardizing:

$$Z = \frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \sim N(0, 1).$$

Step 2: Chi-square quantity independent of Z . By Theorem 5.2, $V = \text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ and is independent of $\hat{\beta}$. Since Z is a (deterministic) function of $\hat{\beta}_j$, which is a coordinate of $\hat{\beta}$, and a function of an independent random variable remains independent, Z is independent of V .

Step 3. By definition, if $Z \sim N(0, 1)$ and $V \sim \chi_\nu^2$ are independent, then $Z/\sqrt{V/\nu} \sim t_\nu$. Applying this:

$$\begin{aligned} \frac{Z}{\sqrt{V/(n-k-1)}} &= \frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \cdot \frac{1}{\sqrt{\text{SSE}/[\sigma^2(n-k-1)]}} \\ &= \frac{\hat{\beta}_j - \beta_j}{\sqrt{\text{MSE}} \cdot \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} = \frac{\hat{\beta}_j - \beta_j}{s \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \sim t_{n-k-1}. \end{aligned}$$

Under $H_0: \beta_j = \beta_j^0$, replace β_j by β_j^0 . □

7.2 Overall F Test

Lemma 7.2 (Distribution of SSR/σ^2 Under H_0). Let $\mathbf{J} = \frac{1}{n}\mathbf{1}\mathbf{1}'$ be the projection onto $\mathcal{C}(\mathbf{1})$ of Definition 6.1. Under $H_0: \beta_1 = \dots = \beta_k = 0$ (with (A1)–(A5)):

$$\frac{\text{SSR}}{\sigma^2} \sim \chi_k^2, \quad \text{and SSR is independent of SSE.}$$

Proof. Step 1: Express SSR/σ^2 as a quadratic form in standard normals. Two matrix observations are useful:

- $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ by definition of the hat matrix.
- $\bar{\mathbf{Y}}\mathbf{1} = \mathbf{J}\mathbf{Y}$, because $\mathbf{J}\mathbf{Y} = n^{-1}\mathbf{1}(\mathbf{1}'\mathbf{Y}) = n^{-1}\mathbf{1} \cdot n\bar{Y} = \bar{Y}\mathbf{1}$.

Therefore

$$\hat{\mathbf{Y}} - \bar{\mathbf{Y}}\mathbf{1} = \mathbf{H}\mathbf{Y} - \mathbf{J}\mathbf{Y} = (\mathbf{H} - \mathbf{J})\mathbf{Y},$$

$$\text{and } \text{SSR} = (\hat{\mathbf{Y}} - \bar{\mathbf{Y}}\mathbf{1})'(\hat{\mathbf{Y}} - \bar{\mathbf{Y}}\mathbf{1}) = \mathbf{Y}'(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J})\mathbf{Y}.$$

Under H_0 , $\boldsymbol{\beta} = (\beta_0, 0, \dots, 0)'$, so $\mathbf{X}\boldsymbol{\beta} = \beta_0\mathbf{1}$ and $\mathbf{Y} = \beta_0\mathbf{1} + \boldsymbol{\varepsilon}$. We claim $(\mathbf{H} - \mathbf{J})\mathbf{Y} = (\mathbf{H} - \mathbf{J})\boldsymbol{\varepsilon}$: indeed, since $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ we have $\mathbf{H}\mathbf{1} = \mathbf{1}$ (by Theorem 2.9(ii) applied to the first column of $\mathbf{X}$), and $\mathbf{J}\mathbf{1} = n^{-1}\mathbf{1}(\mathbf{1}'\mathbf{1}) = \mathbf{1}$, so

$$(\mathbf{H} - \mathbf{J})(\beta_0\mathbf{1}) = \beta_0(\mathbf{H}\mathbf{1} - \mathbf{J}\mathbf{1}) = \beta_0(\mathbf{1} - \mathbf{1}) = \mathbf{0},$$

$$\text{and hence } (\mathbf{H} - \mathbf{J})\mathbf{Y} = (\mathbf{H} - \mathbf{J})(\beta_0\mathbf{1} + \boldsymbol{\varepsilon}) = (\mathbf{H} - \mathbf{J})\boldsymbol{\varepsilon}.$$

Let $\mathbf{z} = \boldsymbol{\varepsilon}/\sigma$. Under (A5), $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$. Then

$$\frac{\text{SSR}}{\sigma^2} = \frac{\boldsymbol{\varepsilon}'(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J})\boldsymbol{\varepsilon}}{\sigma^2} = \mathbf{z}'(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J})\mathbf{z}.$$

This reduces to $\mathbf{z}'(\mathbf{H} - \mathbf{J})\mathbf{z}$ once we establish in Step 2 that $\mathbf{H} - \mathbf{J}$ is symmetric idempotent, which implies $(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J}) = (\mathbf{H} - \mathbf{J})^2 = \mathbf{H} - \mathbf{J}$.

Step 2: $\mathbf{H} - \mathbf{J}$ is symmetric idempotent with rank k . Symmetry is immediate. For idempotency, we first show

$$\mathbf{H}\mathbf{J} = \mathbf{J} = \mathbf{J}\mathbf{H}. \tag{33}$$

Using $\mathbf{H}\mathbf{1} = \mathbf{1}$: $\mathbf{H}\mathbf{J} = n^{-1}\mathbf{H}\mathbf{1}\mathbf{1}' = n^{-1}\mathbf{1}\mathbf{1}' = \mathbf{J}$; by symmetry, $\mathbf{J}\mathbf{H} = (\mathbf{H}\mathbf{J})' = \mathbf{J}$. Also $\mathbf{J}^2 = n^{-2}\mathbf{1}(\mathbf{1}'\mathbf{1})\mathbf{1}' = \mathbf{J}$. Then

$$(\mathbf{H} - \mathbf{J})^2 = \mathbf{H}^2 - \mathbf{H}\mathbf{J} - \mathbf{J}\mathbf{H} + \mathbf{J}^2 = \mathbf{H} - \mathbf{J} - \mathbf{J} + \mathbf{J} = \mathbf{H} - \mathbf{J}.$$

For the rank, Lemma 2.8 gives

$$\text{rank}(\mathbf{H} - \mathbf{J}) = \text{tr}(\mathbf{H}) - \text{tr}(\mathbf{J}) = (k + 1) - 1 = k,$$

with $\text{tr}(\mathbf{J}) = n^{-1} \text{tr}(\mathbf{1}\mathbf{1}') = 1$.

The Step 1 expression simplifies: $(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J}) = \mathbf{H} - \mathbf{J}$, so $\text{SSR}/\sigma^2 = \mathbf{z}'(\mathbf{H} - \mathbf{J})\mathbf{z} \sim \chi_k^2$ by Theorem 4.5.

Step 3: Independence of SSR and SSE. Apply Theorem 4.6 to $\mathbf{A}_1 = \mathbf{H} - \mathbf{J}$ and $\mathbf{A}_2 = \mathbf{M}$. The required $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$ follows from $\mathbf{H}\mathbf{M} = \mathbf{0}$ (Theorem 2.9(iii)) and $\mathbf{J}\mathbf{M} = \mathbf{J} - \mathbf{J}\mathbf{H} = \mathbf{0}$ by (33):

$$(\mathbf{H} - \mathbf{J})\mathbf{M} = \mathbf{0}.$$

Hence $\text{SSR}/\sigma^2 = \mathbf{z}'(\mathbf{H} - \mathbf{J})\mathbf{z}$ and $\text{SSE}/\sigma^2 = \mathbf{z}'\mathbf{M}\mathbf{z}$ are independent, and so are SSR and SSE. $\square$

Theorem 7.3 (Overall F Test). Under (A1)–(A5), for testing $H_0: \beta_1 = \beta_2 = \dots = \beta_k = 0$ (the third expression below presupposes $\text{SST} > 0$, so that R^2 is defined; under (A5) this holds with probability one, see the end of the proof):

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{\text{SSR}/k}{\text{SSE}/(n-k-1)} = \frac{R^2/k}{(1-R^2)/(n-k-1)} \sim F_{k,n-k-1} \quad \text{under } H_0. \quad (34)$$

Proof. By Lemma 7.2 and Theorem 5.2, under H_0 : $\text{SSR}/\sigma^2 \sim \chi_k^2$, $\text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$, and they are independent. By the definition of the F distribution (fact (N5)):

$$F = \frac{(\text{SSR}/\sigma^2)/k}{(\text{SSE}/\sigma^2)/(n-k-1)} = \frac{\text{SSR}/k}{\text{SSE}/(n-k-1)} = \frac{\text{MSR}}{\text{MSE}} \sim F_{k,n-k-1}.$$

The equivalence $F = \frac{R^2/k}{(1-R^2)/(n-k-1)}$ follows from $R^2 = \text{SSR}/\text{SST}$ and $1 - R^2 = \text{SSE}/\text{SST}$, which require $\text{SST} > 0$ (Definition 6.6). Under (A5) this holds with probability one: pick a unit vector $\mathbf{v} \perp \mathbf{1}$ (one exists since $n \geq 2$), so that $\mathbf{M}_0\mathbf{v} = \mathbf{v}$ and hence $\mathbf{v}'\mathbf{M}_0 = \mathbf{v}'$; if $\text{SST} = \|\mathbf{M}_0\mathbf{Y}\|^2 = 0$ then $\mathbf{v}'\mathbf{Y} = \mathbf{v}'\mathbf{M}_0\mathbf{Y} = 0$, whereas $\mathbf{v}'\mathbf{Y} \sim N(\mathbf{v}'\mathbf{X}\boldsymbol{\beta}, \sigma^2)$ (Theorem 4.2 applied to $\mathbf{Y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, as in the proof of Theorem 5.3) with $\sigma^2 > 0$ by (A3) is a continuous distribution, so $P(\mathbf{v}'\mathbf{Y} = 0) = 0$. $\square$

7.3 General Linear Hypothesis

Theorem 7.4 (General F Test). Under (A1)–(A5), consider $H_0: \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ where $\mathbf{R}$ is a $q \times (k+1)$ matrix of rank q and $\mathbf{r} \in \mathbb{R}^q$. The test statistic

$$F = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{q \cdot s^2} \sim F_{q,n-k-1} \quad \text{under } H_0. \quad (35)$$

Proof. Step 1. By Theorem 5.1, $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$. By Theorem 4.2:

$$\mathbf{R}\hat{\boldsymbol{\beta}} \sim N(\mathbf{R}\boldsymbol{\beta}, \sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}').$$

Under H_0 , $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$, so $\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r} \sim N(\mathbf{0}, \sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}')$.

Step 2: A quadratic form in $\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}$ is chi-square. Let $\boldsymbol{\Sigma}_R = \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$. $\boldsymbol{\Sigma}_R$ is positive definite: for nonzero $\mathbf{v} \in \mathbb{R}^q$,

$$\mathbf{v}'\boldsymbol{\Sigma}_R\mathbf{v} = (\mathbf{R}'\mathbf{v})'(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{R}'\mathbf{v}) > 0,$$

since $\text{rank}(\mathbf{R}) = q$ makes $\mathbf{R}'\mathbf{v} \neq \mathbf{0}$, and $(\mathbf{X}'\mathbf{X})^{-1}$ is positive definite (Theorem 2.4, Step 4). Let $\boldsymbol{\Sigma}_R^{1/2}$ be its positive definite symmetric square root.

Set $\mathbf{w} = \sigma^{-1}\boldsymbol{\Sigma}_R^{-1/2}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})$. Theorem 4.2 gives $\mathbf{w} \sim N(\mathbf{0}, \mathbf{I}_q)$, so $\mathbf{w}'\mathbf{w} \sim \chi_q^2$ by Theorem 4.5:

$$W := \mathbf{w}'\mathbf{w} = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\boldsymbol{\Sigma}_R^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{\sigma^2} \sim \chi_q^2.$$

Step 3. By Theorem 5.2, $V = \text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$, independent of $\hat{\beta}$; since W is a function of $\hat{\beta}$, W is independent of V . Then:

$$F = \frac{W/q}{V/(n-k-1)} = \frac{(\mathbf{R}\hat{\beta} - \mathbf{r})'\Sigma_R^{-1}(\mathbf{R}\hat{\beta} - \mathbf{r})/(q\sigma^2)}{\text{SSE}/[\sigma^2(n-k-1)]} = \frac{(\mathbf{R}\hat{\beta} - \mathbf{r})'\Sigma_R^{-1}(\mathbf{R}\hat{\beta} - \mathbf{r})}{q \cdot s^2} \sim F_{q, n-k-1}. \quad \square$$

Remark 7.5 (Special cases). The individual t test (Theorem 7.1) is the case $q = 1$, $\mathbf{R} = \mathbf{e}_j'$ (the transpose of the j -th standard basis vector), $\mathbf{r} = \beta_j^0$, where $t^2 = F \sim F_{1, n-k-1}$. The overall F test (Theorem 7.3) is the case $\mathbf{R} = [\mathbf{0}_{k \times 1} \mid \mathbf{I}_k]$ (of rank $q = k$), $\mathbf{r} = \mathbf{0}_k$: by Corollary 7.6 below, the statistic (35) then equals $\frac{(\text{SSE}_R - \text{SSE}_U)/k}{\text{SSE}_U/(n-k-1)}$, where the reduced model minimizes $\|\mathbf{Y} - \mathbf{X}\beta\|^2$ over $\beta = (\beta_0, 0, \dots, 0)'$, i.e. $\|\mathbf{Y} - \beta_0 \mathbf{1}\|^2$ over β_0 ; its normal equation $\mathbf{1}'\mathbf{1}\beta_0 = \mathbf{1}'\mathbf{Y}$ (Theorem 2.4 with $\mathbf{X} = \mathbf{1}$) gives $\hat{\beta}_0 = \bar{Y}$, so the reduced fit is $\bar{Y}\mathbf{1}$, $\text{SSE}_R = \|\mathbf{Y} - \bar{Y}\mathbf{1}\|^2 = \text{SST}$, and $\text{SSE}_R - \text{SSE}_U = \text{SST} - \text{SSE} = \text{SSR}$ by Theorem 6.4. The statistic is therefore $(\text{SSR}/k)/(\text{SSE}/(n-k-1)) = \text{MSR}/\text{MSE}$, as in (34).

Corollary 7.6 (Nested F Test via Residual Sum of Squares). Let SSE_U and SSE_R be the residual sums of squares from the full and reduced models under $H_0: \mathbf{R}\beta = \mathbf{r}$ (q constraints, $\text{rank}(\mathbf{R}) = q$). Then the F statistic in Theorem 7.4 is equivalently expressed as

$$F = \frac{(\text{SSE}_R - \text{SSE}_U)/q}{\text{SSE}_U/(n-k-1)}. \quad (36)$$

(This is an algebraic identity between the two expressions for F : like Theorem 10.3, on which its proof rests, it uses only (A4) and $\text{rank}(\mathbf{R}) = q$. Assumptions (A1)–(A5) enter only through the distribution $F \sim F_{q, n-k-1}$ under H_0 in Theorem 7.4.)

Proof. By Theorem 10.3, the constrained estimator satisfies

$$\hat{\beta} - \hat{\beta}_R = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\beta} - \mathbf{r}). \quad (37)$$

Step 1: Decompose SSE_R . Since $\mathbf{Y} - \mathbf{X}\hat{\beta}_R = (\mathbf{Y} - \mathbf{X}\hat{\beta}) + \mathbf{X}(\hat{\beta} - \hat{\beta}_R) = \mathbf{e} + \mathbf{X}(\hat{\beta} - \hat{\beta}_R)$,

$$\begin{aligned} \text{SSE}_R &= \|\mathbf{Y} - \mathbf{X}\hat{\beta}_R\|^2 = \|\mathbf{e}\|^2 + \|\mathbf{X}(\hat{\beta} - \hat{\beta}_R)\|^2 + 2\mathbf{e}'\mathbf{X}(\hat{\beta} - \hat{\beta}_R) \\ &= \text{SSE}_U + \|\mathbf{X}(\hat{\beta} - \hat{\beta}_R)\|^2, \end{aligned}$$

since the cross term vanishes: $\mathbf{e}'\mathbf{X} = \mathbf{0}'$ by Theorem 2.12(i).

Step 2: Simplify the squared norm. Let $\mathbf{a} = [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\beta} - \mathbf{r})$ denote the trailing factor (a $q \times 1$ vector) in (37), so that $\hat{\beta} - \hat{\beta}_R = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{a}$. Then

$$\begin{aligned} \|\mathbf{X}(\hat{\beta} - \hat{\beta}_R)\|^2 &= (\hat{\beta} - \hat{\beta}_R)'\mathbf{X}'\mathbf{X}(\hat{\beta} - \hat{\beta}_R) \\ &= [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{a}]'\mathbf{X}'\mathbf{X}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{a}] \\ &= \mathbf{a}'\mathbf{R} \underbrace{(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}}_{=(\mathbf{X}'\mathbf{X})^{-1}} \mathbf{R}'\mathbf{a} \\ &= \mathbf{a}'\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{a}. \end{aligned}$$

Expanding $\mathbf{a}$:

$$\|\mathbf{X}(\hat{\beta} - \hat{\beta}_R)\|^2 = (\mathbf{R}\hat{\beta} - \mathbf{r})' \underbrace{[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}}_{=\mathbf{I}_q} (\mathbf{R}\hat{\beta} - \mathbf{r}).$$

The bracketed product collapses to $\mathbf{I}_q$, leaving

$$\text{SSE}_R - \text{SSE}_U = \|\mathbf{X}(\hat{\beta} - \hat{\beta}_R)\|^2 = (\mathbf{R}\hat{\beta} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\beta} - \mathbf{r}). \quad (38)$$

Step 3: Compare with Theorem 7.4. The numerator in Theorem 7.4's F statistic equals the right-hand side of (38) divided by q . Since the denominator $s^2 = \text{SSE}_U/(n - k - 1)$ is the same, substituting yields

$$F = \frac{(\text{SSE}_R - \text{SSE}_U)/q}{\text{SSE}_U/(n - k - 1)}. \quad \square$$

Remark 7.7 (Logical status of the forward reference). The proof relies on the constrained estimator formula (37) from Theorem 10.3 in Section 10; Theorem 10.3 is purely algebraic and does not depend on anything in this section, so no circularity arises. A reader seeking strict linearity can defer this corollary until after Section 10.

In the special case of *homogeneous* constraints, $\mathbf{r} = \mathbf{0}$, there is also a derivation without constrained OLS: the feasible set $\{\boldsymbol{\beta} : \mathbf{R}\boldsymbol{\beta} = \mathbf{0}\}$ is then a subspace, $\mathbf{X}\{\boldsymbol{\beta} : \mathbf{R}\boldsymbol{\beta} = \mathbf{0}\}$ is a subspace of $\mathcal{C}(\mathbf{X})$ of dimension $k + 1 - q$ (by (L1) and (A4)), and SSE_R is the squared distance from $\mathbf{Y}$ to it. Writing $\mathbf{H}_R$ for the orthogonal projection onto this subspace (Remark 2.10; it is symmetric idempotent with column space the subspace), $\mathbf{H}\mathbf{H}_R = \mathbf{H}_R$ (the subspace lies in $\mathcal{C}(\mathbf{X})$, on which $\mathbf{H}$ acts as the identity), so $\mathbf{H} - \mathbf{H}_R$ is symmetric idempotent of rank q , $\mathbf{Y} = \mathbf{H}_R\mathbf{Y} + (\mathbf{H} - \mathbf{H}_R)\mathbf{Y} + \mathbf{M}\mathbf{Y}$ is an orthogonal decomposition, and $\text{SSE}_R - \text{SSE}_U = \mathbf{Y}'(\mathbf{H} - \mathbf{H}_R)\mathbf{Y}$ by Pythagoras; Theorem 10.3 with $\mathbf{r} = \mathbf{0}$ gives the explicit form $\mathbf{H}_R = \mathbf{H} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. For $\mathbf{r} \neq \mathbf{0}$ the feasible set is an affine set, not a subspace, and this identity is *false*: $\mathbf{Y}'(\mathbf{H} - \mathbf{H}_R)\mathbf{Y}$ with the same $\mathbf{H}_R$ equals $(\mathbf{R}\hat{\boldsymbol{\beta}})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}})$, whereas the correct value (38) is centred at $\mathbf{r}$. The constrained-OLS route of the proof above covers general $\mathbf{r}$.

Corollary 7.8 ($t^2 = F$ for a Single Linear Constraint). For testing $H_0: \mathbf{c}'\boldsymbol{\beta} = r_0$ (a single constraint, $q = 1$, $\mathbf{R} = \mathbf{c}'$ with $\mathbf{c} \neq \mathbf{0}$), the F statistic from Theorem 7.4 equals the square of the t statistic:

$$t = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}}, \quad F = t^2,$$

and $F = t^2 \sim F_{1, n-k-1}$ under H_0 by Theorem 7.4. The identity $F = t^2$ itself is algebraic: like Corollary 7.6, it uses only (A4) (through $\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c} > 0$ for $\mathbf{c} \neq \mathbf{0}$, Theorem 2.4, Step 4), and (A1)–(A5) enter only through the distribution.

Proof. With $q = 1$, $\mathbf{R} = \mathbf{c}'$, $\mathbf{r} = r_0$, the general F statistic becomes:

$$F = \frac{(\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0)[\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}]^{-1}(\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0)}{1 \cdot s^2} = \left(\frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \right)^2 = t^2. \quad \square$$

7.4 Confidence Intervals and Regions

Theorem 7.9 (Confidence Interval for $\mathbf{c}'\boldsymbol{\beta}$). Under (A1)–(A5), for $\mathbf{c} \neq \mathbf{0}$ (so that $\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c} > 0$ by Theorem 2.4, Step 4), a $(1 - \alpha) \times 100\%$ confidence interval for $\mathbf{c}'\boldsymbol{\beta}$ is

$$\mathbf{c}'\hat{\boldsymbol{\beta}} \pm t_{\alpha/2, n-k-1} \cdot s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}, \quad (39)$$

where $t_{\alpha/2, n-k-1}$ is the upper $\alpha/2$ quantile of t_{n-k-1} .

Proof. By Theorem 4.2 applied to $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$ with $\mathbf{A} = \mathbf{c}'$ (a $1 \times (k + 1)$ row vector), $\mathbf{c}'\hat{\boldsymbol{\beta}} \sim N(\mathbf{c}'\boldsymbol{\beta}, \sigma^2\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c})$. Standardizing:

$$Z = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{\sigma\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \sim N(0, 1).$$

By Theorem 5.2, $V = \text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ and is independent of $\hat{\boldsymbol{\beta}}$; since Z is a function of $\hat{\boldsymbol{\beta}}$ alone, Z is independent of V . By definition of the t distribution:

$$\frac{Z}{\sqrt{V/(n - k - 1)}} = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{\sigma\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \cdot \frac{1}{\sqrt{\text{SSE}/[\sigma^2(n - k - 1)]}} = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \sim t_{n-k-1},$$

where the last step uses $s^2 = \text{SSE}/(n - k - 1)$. The pivot gives, by the symmetry of the t distribution about 0, equation (18),

$$\mathbb{P}\left(-t_{\alpha/2} \leq \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \leq t_{\alpha/2}\right) = 1 - \alpha,$$

and rearranging for $\mathbf{c}'\boldsymbol{\beta}$ yields the interval. $\square$

Theorem 7.10 (Joint Confidence Ellipsoid). Under (A1)–(A5), a $(1 - \alpha) \times 100\%$ joint confidence region for $\boldsymbol{\beta}$ is

$$\{\boldsymbol{\beta} \in \mathbb{R}^{k+1} : (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \leq (k + 1)s^2 F_{\alpha, k+1, n-k-1}\}. \quad (40)$$

Proof. Step 1: A chi-square quadratic form in $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$. By Theorem 5.1,

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \sim N(\mathbf{0}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}).$$

Let $(\mathbf{X}'\mathbf{X})^{1/2}$ be the positive definite symmetric square root and define $\mathbf{w} = \sigma^{-1}(\mathbf{X}'\mathbf{X})^{1/2}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$. Theorem 4.2 gives $\mathbf{w} \sim N(\mathbf{0}, \mathbf{I}_{k+1})$, so $\mathbf{w}'\mathbf{w} \sim \chi_{k+1}^2$ by Theorem 4.5:

$$\mathbf{w}'\mathbf{w} = \frac{(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})}{\sigma^2} \sim \chi_{k+1}^2.$$

Step 2: Form the F ratio. This is a function of $\hat{\boldsymbol{\beta}}$, hence independent of $\text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ (Theorem 5.2). By definition of F ,

$$\frac{(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/[(k + 1)\sigma^2]}{\text{SSE}/[\sigma^2(n - k - 1)]} = \frac{(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})}{(k + 1)s^2} \sim F_{k+1, n-k-1}.$$

The confidence region collects the $\boldsymbol{\beta}$ for which this statistic is at most $F_{\alpha, k+1, n-k-1}$. $\square$

8 Partitioned Regression

Theorem 8.1 is an algebraic identity that uses only the standing assumption (A4); Theorem 8.3 uses (A2) in addition. No distributional assumption is used in this section.

8.1 The Frisch–Waugh–Lovell Theorem

Theorem 8.1 (Frisch–Waugh–Lovell). Partition $\mathbf{X} = [\mathbf{X}_1 \ \mathbf{X}_2]$ and $\boldsymbol{\beta} = (\boldsymbol{\beta}'_1, \boldsymbol{\beta}'_2)'$, so the model is $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$. Let $\mathbf{M}_2 = \mathbf{I} - \mathbf{X}_2(\mathbf{X}'_2\mathbf{X}_2)^{-1}\mathbf{X}'_2$ be the residual maker for $\mathbf{X}_2$. Then:

- (i) $\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}'_1\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{M}_2\mathbf{Y}$.
- (ii) The OLS coefficient vector from regressing $\mathbf{M}_2\mathbf{Y}$ on $\mathbf{M}_2\mathbf{X}_1$ is the $\hat{\boldsymbol{\beta}}_1$ of (i), and the residual vector of that regression equals the residual vector $\mathbf{e}$ from the full regression.

Proof. (i). The normal equations for the full model are:

$$\begin{pmatrix} \mathbf{X}'_1\mathbf{X}_1 & \mathbf{X}'_1\mathbf{X}_2 \\ \mathbf{X}'_2\mathbf{X}_1 & \mathbf{X}'_2\mathbf{X}_2 \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}}_1 \\ \hat{\boldsymbol{\beta}}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{X}'_1\mathbf{Y} \\ \mathbf{X}'_2\mathbf{Y} \end{pmatrix}.$$

The second block equation reads $\mathbf{X}'_2\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}'_2\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \mathbf{X}'_2\mathbf{Y}$, which rearranges to

$$\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}'_2\mathbf{X}_2)^{-1}\mathbf{X}'_2(\mathbf{Y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1).$$

Substituting this into the first block equation $\mathbf{X}'_1\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}'_1\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \mathbf{X}'_1\mathbf{Y}$:

$$\mathbf{X}'_1\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \underbrace{\mathbf{X}'_1\mathbf{X}_2(\mathbf{X}'_2\mathbf{X}_2)^{-1}\mathbf{X}'_2}_{=\mathbf{X}'_1\mathbf{H}_2}(\mathbf{Y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1) = \mathbf{X}'_1\mathbf{Y}, \quad (41)$$

where we recognize the projection onto $\mathcal{C}(\mathbf{X}_2)$:

$$\mathbf{H}_2 := \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2' = \mathbf{I} - \mathbf{M}_2.$$

Expanding (41) using $\mathbf{X}_1'\mathbf{H}_2 = \mathbf{X}_1' - \mathbf{X}_1'\mathbf{M}_2$:

$$\mathbf{X}_1'\mathbf{X}_1\hat{\beta}_1 + \mathbf{X}_1'(\mathbf{Y} - \mathbf{X}_1\hat{\beta}_1) - \mathbf{X}_1'\mathbf{M}_2(\mathbf{Y} - \mathbf{X}_1\hat{\beta}_1) = \mathbf{X}_1'\mathbf{Y}.$$

The first two terms combine to $\mathbf{X}_1'\mathbf{Y}$, cancelling the right-hand side, leaving $\mathbf{X}_1'\mathbf{M}_2(\mathbf{Y} - \mathbf{X}_1\hat{\beta}_1) = \mathbf{0}$, or

$$\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1\hat{\beta}_1 = \mathbf{X}_1'\mathbf{M}_2\mathbf{Y}.$$

Since $\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1 = (\mathbf{M}_2\mathbf{X}_1)'(\mathbf{M}_2\mathbf{X}_1)$ is positive definite (the columns of $\mathbf{M}_2\mathbf{X}_1$ are linearly independent: if $\mathbf{M}_2\mathbf{X}_1\mathbf{v} = \mathbf{0}$ then $\mathbf{X}_1\mathbf{v} = \mathbf{H}_2\mathbf{X}_1\mathbf{v} \in \mathcal{C}(\mathbf{X}_2)$, say $\mathbf{X}_1\mathbf{v} = \mathbf{X}_2\mathbf{w}$, so $[\mathbf{X}_1 \ \mathbf{X}_2](\mathbf{v}', -\mathbf{w}')' = \mathbf{0}$ and (A4) forces $\mathbf{v} = \mathbf{0}$):

$$\hat{\beta}_1 = (\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{M}_2\mathbf{Y}.$$

(ii). Let $\tilde{\mathbf{Y}} = \mathbf{M}_2\mathbf{Y}$ and $\tilde{\mathbf{X}}_1 = \mathbf{M}_2\mathbf{X}_1$. The columns of $\tilde{\mathbf{X}}_1$ are linearly independent (shown in (i)), so the OLS coefficient vector from regressing $\tilde{\mathbf{Y}}$ on $\tilde{\mathbf{X}}_1$ is defined, and, using $\mathbf{M}_2' = \mathbf{M}_2 = \mathbf{M}_2^2$ (Theorem 2.9(i) for the design $\mathbf{X}_2$, which has full column rank by (A4)) and then (i),

$$(\tilde{\mathbf{X}}_1'\tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1'\tilde{\mathbf{Y}} = (\mathbf{X}_1'\mathbf{M}_2'\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{M}_2'\mathbf{M}_2\mathbf{Y} = (\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{M}_2\mathbf{Y} = \hat{\beta}_1.$$

The residual from this regression is therefore

$$\tilde{\mathbf{e}} = \mathbf{M}_2\mathbf{Y} - \mathbf{M}_2\mathbf{X}_1\hat{\beta}_1 = \mathbf{M}_2(\mathbf{Y} - \mathbf{X}_1\hat{\beta}_1).$$

Substituting $\mathbf{Y} - \mathbf{X}_1\hat{\beta}_1 = \mathbf{X}_2\hat{\beta}_2 + \mathbf{e}$ from the full regression and using $\mathbf{M}_2\mathbf{X}_2 = \mathbf{0}$:

$$\tilde{\mathbf{e}} = \mathbf{M}_2\mathbf{X}_2\hat{\beta}_2 + \mathbf{M}_2\mathbf{e} = \mathbf{M}_2\mathbf{e}. \quad (42)$$

Finally, $\mathbf{M}_2\mathbf{e} = \mathbf{e}$: since $\mathbf{X}'\mathbf{e} = \mathbf{0}$ (Theorem 2.12(i)) includes $\mathbf{X}_2'\mathbf{e} = \mathbf{0}$,

$$\mathbf{M}_2\mathbf{e} = \mathbf{e} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{e} = \mathbf{e}.$$

Hence $\tilde{\mathbf{e}} = \mathbf{e}$. □

Remark 8.2 (The key identity $\mathbf{M}_2\mathbf{M} = \mathbf{M}$). The computation above rests on the identity $\mathbf{M}_2\mathbf{e} = \mathbf{e}$, which can be written equivalently as $\mathbf{M}_2\mathbf{M} = \mathbf{M}$ (applying both sides to $\mathbf{Y}$). This holds because $\mathcal{C}(\mathbf{X}_2) \subseteq \mathcal{C}(\mathbf{X})$, so the orthogonal complement $\mathcal{C}(\mathbf{X})^\perp \subseteq \mathcal{C}(\mathbf{X}_2)^\perp$: anything already orthogonal to $\mathbf{X}$ is automatically orthogonal to $\mathbf{X}_2$, and $\mathbf{M}_2$ leaves it unchanged. This identity is the abstract reason why “residualizing with respect to a subset of regressors” preserves the full-model residual.

8.2 Omitted Variable Bias

Theorem 8.3 (Omitted Variable Bias). Suppose the true model is $\mathbf{Y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \boldsymbol{\varepsilon}$ with $E[\boldsymbol{\varepsilon}] = \mathbf{0}$ (assumption (A2)), but the researcher estimates the “short” regression $\mathbf{Y} = \mathbf{X}_1\gamma + \mathbf{u}$, obtaining $\tilde{\gamma} = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{Y}$. Then:

$$E[\tilde{\gamma}] = \beta_1 + \underbrace{(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2}_{\text{auxiliary regression coefficients}} \beta_2. \quad (43)$$

The estimator $\tilde{\gamma}$ is unbiased for β_1 if and only if $\mathbf{X}_1'\mathbf{X}_2\beta_2 = \mathbf{0}$, which holds when either $\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$ ($\mathbf{X}_1$ and $\mathbf{X}_2$ are orthogonal) or $\beta_2 = \mathbf{0}$ (the omitted variables are irrelevant).

Proof. Substituting the true DGP $\mathbf{Y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \boldsymbol{\varepsilon}$ into the short-regression estimator:

$$\begin{aligned} \tilde{\gamma} &= (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{Y} = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'(\mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \boldsymbol{\varepsilon}) \\ &= \beta_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\beta_2 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\boldsymbol{\varepsilon}. \end{aligned}$$

Taking expectations (using $E[\boldsymbol{\varepsilon}] = \mathbf{0}$):

$$E[\tilde{\gamma}] = \beta_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\beta_2. \quad \square$$

Remark 8.4 (Two-factor structure of the bias). The bias term $(\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{X}_2\boldsymbol{\beta}_2$ is the product of two conceptually distinct quantities:

- $(\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{X}_2$ is the matrix of OLS coefficients from regressing each column of $\mathbf{X}_2$ on $\mathbf{X}_1$ —an *auxiliary regression* describing how the omitted variables relate to the included ones.
- $\boldsymbol{\beta}_2$ is the effect of the omitted variables on $\mathbf{Y}$ in the true model.

The bias vanishes if either factor is zero: if $\mathbf{X}_1$ and $\mathbf{X}_2$ are orthogonal ($\mathbf{X}'_1\mathbf{X}_2 = \mathbf{0}$), omission is harmless; if the omitted variables have no effect ($\boldsymbol{\beta}_2 = \mathbf{0}$), there is nothing to omit. Both ingredients must be nonzero to generate bias.

Remark 8.5 (Connection to FWL). Theorem 8.1 shows that the correct (full-model) coefficient $\hat{\boldsymbol{\beta}}_1$ is obtained by regressing $\mathbf{M}_2\mathbf{Y}$ on $\mathbf{M}_2\mathbf{X}_1$ —that is, after partialling $\mathbf{X}_2$ out of both $\mathbf{Y}$ and $\mathbf{X}_1$. The short regression instead regresses $\mathbf{Y}$ directly on $\mathbf{X}_1$, effectively replacing $\mathbf{M}_2$ with $\mathbf{I}$. The two coefficient vectors differ by an exact sample identity: the first block of the normal equations of the full model, $\mathbf{X}'_1\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}'_1\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \mathbf{X}'_1\mathbf{Y} = \mathbf{X}'_1\mathbf{X}_1\tilde{\boldsymbol{\gamma}}$, gives

$$\tilde{\boldsymbol{\gamma}} - \hat{\boldsymbol{\beta}}_1 = (\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{X}_2\hat{\boldsymbol{\beta}}_2,$$

which has the same form as the bias term in (43) with $\hat{\boldsymbol{\beta}}_2$ in place of $\boldsymbol{\beta}_2$; taking expectations, with $\mathbb{E}[\hat{\boldsymbol{\beta}}_1] = \boldsymbol{\beta}_1$ and $\mathbb{E}[\hat{\boldsymbol{\beta}}_2] = \boldsymbol{\beta}_2$ (Theorem 3.3, under (A2)), recovers (43). OVB measures the cost of not partialling out the omitted variables.

9 Prediction

Let $\mathbf{x}_0 \in \mathbb{R}^{k+1}$ be a new observation's regressor vector. The predicted value is $\hat{Y}_0 = \mathbf{x}'_0\hat{\boldsymbol{\beta}}$.

Theorem 9.1 (Confidence Interval for the Conditional Mean). Under (A1)–(A5), a $(1 - \alpha) \times 100\%$ confidence interval for $\mathbb{E}[Y_0 | \mathbf{x}_0] = \mathbf{x}'_0\boldsymbol{\beta}$ is

$$\mathbf{x}'_0\hat{\boldsymbol{\beta}} \pm t_{\alpha/2, n-k-1} \cdot s\sqrt{\mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}. \quad (44)$$

Proof. This is the special case $\mathbf{c} = \mathbf{x}_0$ of Theorem 7.9; its hypothesis $\mathbf{c} \neq \mathbf{0}$ holds because the first component of $\mathbf{x}_0$ is the intercept's 1 (Definition 1.1). $\square$

Theorem 9.2 (Prediction Interval for a New Observation). Under (A1)–(A5), let $Y_0 = \mathbf{x}'_0\boldsymbol{\beta} + \varepsilon_0$ be a new observation whose error $\varepsilon_0 \sim N(0, \sigma^2)$ is independent of $\boldsymbol{\varepsilon}$ (that is, (A3) and (A5) extend to the new observation). Then a $(1 - \alpha) \times 100\%$ prediction interval for Y_0 is

$$\mathbf{x}'_0\hat{\boldsymbol{\beta}} \pm t_{\alpha/2, n-k-1} \cdot s\sqrt{1 + \mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}. \quad (45)$$

Proof. Step 1: Set up. The prediction error is

$$\hat{Y}_0 - Y_0 = \mathbf{x}'_0(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - \varepsilon_0.$$

Here $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$ depends on the training-sample errors $\boldsymbol{\varepsilon}$, whereas ε_0 is the new-observation error, which by hypothesis is independent of $\boldsymbol{\varepsilon}$; hence $\varepsilon_0 \perp\!\!\!\perp \mathbf{x}'_0(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ (fact (N3)). Moreover the stacked vector $(\boldsymbol{\varepsilon}', \varepsilon_0)'$ is jointly normal: by independence its MGF factorizes as $\mathbb{E}[e^{\mathbf{t}'\boldsymbol{\varepsilon}}]\mathbb{E}[e^{t_0\varepsilon_0}] = \exp(\frac{\sigma^2}{2}\mathbf{t}'\mathbf{t})\exp(\frac{\sigma^2}{2}t_0^2)$, which is the MGF (17) of $N(\mathbf{0}, \sigma^2\mathbf{I}_{n+1})$ (fact (N2) and Definition 4.1). Consequently

$$\begin{pmatrix} \hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \\ \mathbf{e} \\ \varepsilon_0 \end{pmatrix} = \begin{pmatrix} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' & \mathbf{0} \\ \mathbf{M} & \mathbf{0} \\ \mathbf{0}' & 1 \end{pmatrix} \begin{pmatrix} \boldsymbol{\varepsilon} \\ \varepsilon_0 \end{pmatrix} \quad (46)$$

is jointly normal (Theorem 4.2), with $\text{cov}(\mathbf{e}, \hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \sigma^2\mathbf{M}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{0}$ (as in Theorem 5.2, Step 3) and $\text{cov}(\mathbf{e}, \varepsilon_0) = [\mathbf{M} \ \mathbf{0}] \sigma^2\mathbf{I}_{n+1} [\mathbf{0}' \ 1]' = \mathbf{0}$ (Lemma 3.2(ii)).

Step 2: Compute the variance. Using Step 1's independence (so that the variances of the two terms add, fact (N7)) and Lemma 3.2 (with $\mathbf{A} = \mathbf{x}'_0$):

$$\begin{aligned}\text{var}(\hat{Y}_0 - Y_0) &= \text{var}(\mathbf{x}'_0(\hat{\beta} - \beta)) + \text{var}(\varepsilon_0) \\ &= \mathbf{x}'_0 \text{var}(\hat{\beta}) \mathbf{x}_0 + \sigma^2 \\ &= \sigma^2 \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0 + \sigma^2 \\ &= \sigma^2 (1 + \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0).\end{aligned}$$

Step 3: Derive the t pivotal quantity. As a linear function of the jointly normal vector $(\varepsilon', \varepsilon_0)'$ of Step 1, namely $\hat{Y}_0 - Y_0 = [\mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \quad -1] (\varepsilon', \varepsilon_0)'$, the prediction error is normal (Theorem 4.2) with mean 0 and the variance of Step 2:

$$\hat{Y}_0 - Y_0 \sim N(0, \sigma^2(1 + \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0)).$$

For independence from SSE, note that “ $\hat{\beta} \perp\!\!\!\perp \text{SSE}$ and $\varepsilon_0 \perp\!\!\!\perp \text{SSE}$ ” would *not* by itself imply that the function $\mathbf{x}'_0(\hat{\beta} - \beta) - \varepsilon_0$ of the pair is independent of SSE; what is needed is independence of $\mathbf{e}$ from the pair $(\hat{\beta} - \beta, \varepsilon_0)$ jointly. This follows from (46): the vector there is jointly normal and both cross-covariances of $\mathbf{e}$ with the other two blocks vanish, so Theorem 4.3 (with $\mathbf{U} = \mathbf{e}$ and $\mathbf{V} = ((\hat{\beta} - \beta)', \varepsilon_0)'$) gives $\mathbf{e} \perp\!\!\!\perp (\hat{\beta} - \beta, \varepsilon_0)$. Hence $\text{SSE} = \mathbf{e}'\mathbf{e}$ is independent of $\hat{Y}_0 - Y_0$ (fact (N3)). Replacing σ with s gives a t_{n-k-1} pivot (fact (N5), exactly as in Theorem 7.9):

$$\frac{\hat{Y}_0 - Y_0}{s \sqrt{1 + \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0}} \sim t_{n-k-1},$$

and inverting, using (18) exactly as at the end of the proof of Theorem 7.9, yields the prediction interval. $\square$

Remark 9.3. The prediction interval is always wider than the confidence interval for the mean, because the prediction variance includes the additional σ^2 term from the irreducible noise ε_0 .

10 Constrained Least Squares and Modeling Remarks

Theorem 10.3 below is a purely algebraic statement about a given $\mathbf{Y}$ and $\mathbf{X}$; it uses only the standing assumption (A4) and no distributional assumption.

10.1 Remarks on Model Specification

Remark 10.1 (No-intercept model). When the model omits the intercept, $\mathbf{X}$ has no column of ones. The OLS formula $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}$ still applies, but Theorem 2.12(ii) is no longer available (its proof reads $\mathbf{1}'\mathbf{e} = 0$ off the intercept row of $\mathbf{X}'\mathbf{e} = \mathbf{0}$), so $\sum e_i \neq 0$ in general; since Steps 1, 2 and 4 of the proof of Theorem 6.4 all use $\mathbf{1}'\mathbf{e} = 0$, the decomposition $\text{SST} = \text{SSR} + \text{SSE}$ may fail. Consequently the two expressions in Definition 6.6 no longer agree and Proposition 6.7 does not apply: $1 - \text{SSE}/\text{SST}$ can be negative, while $\text{SSR}/\text{SST} \geq 0$ can exceed 1. (For $n = 2$ with the single regressor $\mathbf{X} = (1, 0)'$ and $\mathbf{Y} = (1, 3)'$: $\hat{\beta} = 1$, $\mathbf{e} = (0, 3)'$, $\text{SST} = 2$, $\text{SSR} = 5$, $\text{SSE} = 9$, so $1 - \text{SSE}/\text{SST} = -3.5$ and $\text{SSR}/\text{SST} = 2.5$.) Use the uncentered $R^2 = 1 - \text{SSE}/\sum Y_i^2 = 1 - \text{SSE}/\mathbf{Y}'\mathbf{Y}$ instead (for $\mathbf{Y} \neq \mathbf{0}$): it lies in $[0, 1]$ because $\text{SSE} = \|\mathbf{M}\mathbf{Y}\|^2 = \mathbf{Y}'\mathbf{M}\mathbf{Y}$ and $\mathbf{Y}'\mathbf{Y} - \mathbf{Y}'\mathbf{M}\mathbf{Y} = \mathbf{Y}'\mathbf{H}\mathbf{Y} = \|\mathbf{H}\mathbf{Y}\|^2 \geq 0$ (Theorem 2.9(i)); in the example it equals $1 - 9/10 = 0.1$.

Remark 10.2 (Dummy variable trap). A categorical variable with g levels requires $g - 1$ dummy regressors when the model includes an intercept. Including all g dummies creates the linear dependence $\mathbf{1} = D_1 + \cdots + D_g$, violating (A4).

10.2 Constrained Least Squares

Theorem 10.3 (Constrained OLS). Consider $\min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|^2$ subject to $\mathbf{R}\beta = \mathbf{r}$ (q linear constraints, $\text{rank}(\mathbf{R}) = q$). The constrained estimator is:

$$\hat{\beta}_R = \hat{\beta} - (\mathbf{X}'\mathbf{X})^{-1} \mathbf{R}' [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{R}']^{-1} (\mathbf{R}\hat{\beta} - \mathbf{r}), \quad (47)$$

and $\text{SSE}_R \geq \text{SSE}_U$ (where SSE_U is the unconstrained residual sum of squares).

Proof. The objective $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ is a convex quadratic function of $\boldsymbol{\beta}$ (its Hessian $2\mathbf{X}'\mathbf{X}$ is positive definite by (A4)), and the constraint $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ is affine. Rather than invoking the theory of constrained optimization (the KKT conditions), we use the Lagrangian only to *produce a candidate* (Steps 1–2) and then verify directly that the candidate is the unique minimizer over the feasible set (Step 3); nothing from outside these notes is needed.

Step 1: Lagrangian and first-order conditions. The Lagrangian is $\mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\lambda}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) + 2\boldsymbol{\lambda}'(\mathbf{R}\boldsymbol{\beta} - \mathbf{r})$ (the factor 2 simplifies later algebra). By Lemma 2.2, differentiating $(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{Y}'\mathbf{Y} - 2\mathbf{Y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}$ with respect to $\boldsymbol{\beta}'$ gives $-2\mathbf{Y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}$. The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\beta}'} = -2\mathbf{Y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X} + 2\boldsymbol{\lambda}'\mathbf{R} = \mathbf{0}', \quad (\text{i})$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\lambda}'} = 2(\mathbf{R}\boldsymbol{\beta} - \mathbf{r})' = \mathbf{0}'. \quad (\text{ii})$$

Transposing (i) gives $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}}_R + \mathbf{R}'\hat{\boldsymbol{\lambda}} = \mathbf{X}'\mathbf{Y}$, so

$$\hat{\boldsymbol{\beta}}_R = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}} = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}}. \quad (48)$$

Step 2: Solve for $\hat{\boldsymbol{\lambda}}$. Left-multiply (48) by $\mathbf{R}$ and impose the constraint (ii), $\mathbf{R}\hat{\boldsymbol{\beta}}_R = \mathbf{r}$:

$$\mathbf{r} = \mathbf{R}\hat{\boldsymbol{\beta}}_R = \mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}}.$$

Since $\text{rank}(\mathbf{R}) = q$ and $(\mathbf{X}'\mathbf{X})^{-1}$ is positive definite (Theorem 2.4, Step 4), $\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$ is positive definite (hence invertible); see Step 2 of the proof of Theorem 7.4 for the detailed argument. Rearranging:

$$\hat{\boldsymbol{\lambda}} = [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}).$$

Substituting back into (48) gives (47).

Step 3: $\hat{\boldsymbol{\beta}}_R$ is the unique minimizer over the feasible set. Define $\hat{\boldsymbol{\beta}}_R$ by (47) and $\hat{\boldsymbol{\lambda}}$ as in Step 2. By construction, $\mathbf{R}\hat{\boldsymbol{\beta}}_R = \mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}} = \mathbf{R}\hat{\boldsymbol{\beta}} - (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) = \mathbf{r}$ (feasibility), and, transposing (i),

$$\mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R) = \mathbf{X}'\mathbf{Y} - \mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} + \mathbf{R}'\hat{\boldsymbol{\lambda}} = \mathbf{R}'\hat{\boldsymbol{\lambda}}, \quad (49)$$

using $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}$. Now let $\boldsymbol{\beta}$ be any feasible vector, $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$. Writing $\mathbf{Y} - \mathbf{X}\boldsymbol{\beta} = (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R) + \mathbf{X}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta})$, the cross term vanishes:

$$(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R)'(\mathbf{X}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta})) = \hat{\boldsymbol{\lambda}}'\mathbf{R}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta}) = \hat{\boldsymbol{\lambda}}'(\mathbf{r} - \mathbf{r}) = 0$$

by (49). Hence

$$\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R\|^2 + \|\mathbf{X}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta})\|^2 \geq \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R\|^2,$$

with equality iff $\mathbf{X}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta}) = \mathbf{0}$, i.e. (by (A4)) iff $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}_R$. So $\hat{\boldsymbol{\beta}}_R$ is the unique constrained minimizer and $\text{SSE}_R = \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R\|^2$.

Step 4: $\text{SSE}_R \geq \text{SSE}_U$. The constrained problem minimizes $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ over $\{\boldsymbol{\beta} : \mathbf{R}\boldsymbol{\beta} = \mathbf{r}\} \subseteq \mathbb{R}^{k+1}$. Since $\hat{\boldsymbol{\beta}}$ minimizes the same objective over the full space, the minimum over a subset cannot be smaller: $\text{SSE}_R \geq \text{SSE}_U$. Equality holds iff $\hat{\boldsymbol{\beta}}$ already satisfies the constraint, i.e., $\mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{r}$: if $\mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{r}$, then (47) gives $\hat{\boldsymbol{\beta}}_R = \hat{\boldsymbol{\beta}}$ and hence $\text{SSE}_R = \text{SSE}_U$; conversely, if $\text{SSE}_R = \text{SSE}_U$, then $Q(\hat{\boldsymbol{\beta}}_R) = \text{SSE}_R = \text{SSE}_U = Q(\hat{\boldsymbol{\beta}})$, so $\hat{\boldsymbol{\beta}}_R$ minimizes Q over all of $\mathbb{R}^{k+1}$, hence $\hat{\boldsymbol{\beta}}_R = \hat{\boldsymbol{\beta}}$ by the uniqueness in Theorem 2.4 (by (7) with $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}_R$, $\mathbf{X}(\hat{\boldsymbol{\beta}}_R - \hat{\boldsymbol{\beta}}) = \mathbf{0}$, so $\hat{\boldsymbol{\beta}}_R = \hat{\boldsymbol{\beta}}$ under (A4)), and therefore $\mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{R}\hat{\boldsymbol{\beta}}_R = \mathbf{r}$ by the feasibility established in Step 3. $\square$

11 Exercises

Tier A: Reproduce the Derivation — Estimation, Distribution Theory and the Overall F Test (Sections 2–7)

Exercise A1. Establish the matrix differentiation formulas of Lemma 2.2: for constant $\mathbf{a} \in \mathbb{R}^p$ and constant $\mathbf{A} \in \mathbb{R}^{p \times p}$, prove $D(\mathbf{a}'\boldsymbol{\beta}) = \mathbf{a}'$ and $D(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')$ by computing the partial derivatives $\partial/\partial\beta_j$ coordinate by coordinate. Then apply the formulas to $Q(\boldsymbol{\beta}) = \mathbf{Y}'\mathbf{Y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{Y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}$ to obtain $DQ(\boldsymbol{\beta}) = -2\mathbf{Y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}$.

Exercise A2. Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be symmetric and idempotent ($\mathbf{A}^2 = \mathbf{A} = \mathbf{A}'$). (a) Show every eigenvalue of $\mathbf{A}$ lies in $\{0, 1\}$. (b) Conclude $\text{tr}(\mathbf{A}) = \text{rank}(\mathbf{A})$. (c) Verify the formula on $\mathbf{H}$ and $\mathbf{M}$: $\text{tr}(\mathbf{H}) = k + 1$ and $\text{tr}(\mathbf{M}) = n - k - 1$.

Exercise A3. Derive the OLS estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ by minimizing $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$. Verify the second-order condition.

Exercise A4. For $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and $\mathbf{M} = \mathbf{I} - \mathbf{H}$: (a) prove $\mathbf{H}$ and $\mathbf{M}$ are symmetric and idempotent; (b) prove $\mathbf{H}\mathbf{X} = \mathbf{X}$, $\mathbf{M}\mathbf{X} = \mathbf{0}$, and $\mathbf{H}\mathbf{M} = \mathbf{0}$; (c) conclude $\text{tr}(\mathbf{H}) = k + 1$ and $\text{tr}(\mathbf{M}) = n - k - 1$.

Exercise A5. Prove that $E[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$ under (A2) and (A4), and derive $\text{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ under (A3) and (A4).

Exercise A6. State and prove the Gauss–Markov theorem.

Exercise A7. Prove that $\text{MSE} = \text{SSE}/(n - k - 1)$ is an unbiased estimator of σ^2 under (A2)–(A4).

Exercise A8. Let $\mathbf{z} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. (a) For any constant matrix $\mathbf{A}$ and vector $\mathbf{b}$, prove $\mathbf{A}\mathbf{z} + \mathbf{b} \sim N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$ via the moment generating function. (b) Specialise to $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and let $\mathbf{A}$ be symmetric idempotent of rank r ; prove $\mathbf{z}'\mathbf{A}\mathbf{z} \sim \chi_r^2$ via the spectral decomposition.

Exercise A9. (Independence of quadratic forms.) Let $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and let $\mathbf{A}_1, \mathbf{A}_2$ be symmetric idempotent with $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$. Prove that $\mathbf{z}'\mathbf{A}_1\mathbf{z}$ and $\mathbf{z}'\mathbf{A}_2\mathbf{z}$ are independent. *Hint:* use joint normality of $(\mathbf{A}_1\mathbf{z}, \mathbf{A}_2\mathbf{z})$ and the equivalence “uncorrelated $\Leftrightarrow$ independent” under joint normality.

Exercise A10. Under (A1)–(A5), prove $\text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ and that SSE is independent of $\hat{\boldsymbol{\beta}}$.

Exercise A11. Prove the variance decomposition $\text{SST} = \text{SSR} + \text{SSE}$, and derive the overall F statistic including the proof that $\text{SSR}/\sigma^2 \sim \chi_k^2$ under $H_0: \beta_1 = \cdots = \beta_k = 0$.

Tier B: Reproduce and Specialize — Inference and Prediction (Sections 2, 3, 7, 9; Exercise B5 extends Section 3 to non-spherical errors)

Exercise B1. Let $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. (a) Show $0 \leq h_{ii} \leq 1$. (b) Show $\sum_{i=1}^n h_{ii} = k + 1$. (c) Derive $\text{var}(e_i) = \sigma^2(1 - h_{ii})$.

Exercise B2. Consider testing $H_0: \mathbf{c}'\boldsymbol{\beta} = r_0$ for a given nonzero vector $\mathbf{c}$. (a) Derive the distribution of $\mathbf{c}'\hat{\boldsymbol{\beta}}$. (b) Construct the t statistic and prove $t^2 \sim F_{1, n-k-1}$ under H_0 .

Exercise B3. Consider two nested models: the full model with k regressors and a reduced model with $k_0 < k$ regressors. Let SSE_R and SSE_U denote the respective residual sums of squares. (a) Prove $\text{SSE}_R \geq \text{SSE}_U$. (b) Show that $\frac{(\text{SSE}_R - \text{SSE}_U)/(k - k_0)}{\text{SSE}_U/(n - k - 1)} \sim F_{k-k_0, n-k-1}$ under $H_0: \beta_{k_0+1} = \cdots = \beta_k = 0$, where the reduced model retains the intercept and the first k_0 regressors.

Exercise B4. (General linear hypothesis.) For $H_0: \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ with $\mathbf{R}$ a $q \times (k + 1)$ matrix of rank q , derive the test statistic

$$F = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})/q}{\text{SSE}/(n - k - 1)}$$

and prove $F \sim F_{q,n-k-1}$ under H_0 .

Exercise B5. Assume (A1), (A2) and (A4), but suppose that in place of (A3) we have $\text{var}(\varepsilon) = \sigma^2 \Omega$, where Ω is known, symmetric, and positive definite (but $\neq \mathbf{I}$). (a) Show that OLS is unbiased but derive its actual (incorrect) variance. (b) Define $\hat{\beta}_{\text{GLS}} = (\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\mathbf{X}'\Omega^{-1}\mathbf{Y}$ and prove it is BLUE.

Exercise B6. Let $\hat{Y}_0 = \mathbf{x}_0'\hat{\beta}$ be the predicted value at a new point $\mathbf{x}_0$, where $Y_0 = \mathbf{x}_0'\beta + \varepsilon_0$ with $E[\varepsilon_0] = 0$, $\text{var}(\varepsilon_0) = \sigma^2$ and ε_0 independent of ε . Assume (A1)–(A4). (a) Derive $\text{var}(\hat{Y}_0 - E[Y_0])$. (b) Derive $\text{var}(\hat{Y}_0 - Y_0)$. (c) Explain why (b) always exceeds (a).

Tier C: Reproduce the Derivation — Partitioned Regression, Model Selection and Constraints (Sections 6, 8, 10)

Exercise C1. (Frisch–Waugh–Lovell.) Partition $\mathbf{X} = [\mathbf{X}_1 \ \mathbf{X}_2]$. Let $\mathbf{M}_2 = \mathbf{I} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'$. Prove: (a) $\hat{\beta}_1 = (\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{M}_2\mathbf{Y}$. (b) The full regression residuals equal those from regressing $\mathbf{M}_2\mathbf{Y}$ on $\mathbf{M}_2\mathbf{X}_1$.

Exercise C2. (Omitted variable bias.) The true model is $\mathbf{Y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \varepsilon$, but only $\mathbf{Y} = \mathbf{X}_1\gamma + \mathbf{u}$ is estimated. Show $E[\tilde{\gamma}] = \beta_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\beta_2$. Under what conditions is $\tilde{\gamma}$ unbiased?

Exercise C3. (Monotonicity of R^2 and the adjusted R^2 criterion.) (a) Prove that adding a regressor cannot decrease R^2 . (b) Show that $\bar{R}^2$ increases if and only if the added variable's $|t|$ -statistic exceeds 1.

Exercise C4. (Constrained least squares.) Consider $\min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|^2$ subject to $\mathbf{R}\beta = \mathbf{r}$. (a) Derive the constrained estimator $\hat{\beta}_R$ via Lagrange multipliers. (b) Show

$$\hat{\beta}_R = \hat{\beta} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\beta} - \mathbf{r}).$$

(c) Prove $\text{SSE}_R \geq \text{SSE}_U$.

Tier D: Apply and Extend

The exercises in this tier apply the theory to objects that do not appear in the text: a data set to be worked by hand, situations in which one has to decide which assumption fails, and two small propositions not proved above.

Exercise D1. (A regression by hand.) Five observations of a regressor x and a response Y are

x_i	1	2	3	4	5
Y_i	1	2	2	4	6

Fit the model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ under (A1)–(A5). (a) Compute $\mathbf{X}'\mathbf{X}$, $\mathbf{X}'\mathbf{Y}$, $(\mathbf{X}'\mathbf{X})^{-1}$ and $\hat{\beta}$. (b) Compute the fitted values $\hat{\mathbf{Y}}$ and residuals $\mathbf{e}$, and verify $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$ (Theorem 2.12). (c) Compute SSE and $\text{MSE} = s^2$. (d) Compute the leverages $h_{11}, \dots, h_{55}$ and verify $0 \leq h_{ii} \leq 1$ and $\sum_i h_{ii} = 2$ (Theorem 2.14). (e) Compute SST, SSR, verify $\text{SST} = \text{SSR} + \text{SSE}$ (Theorem 6.4), and compute R^2 and $\bar{R}^2$. (f) Compute the t statistic for $H_0: \beta_1 = 0$ and the overall F statistic, and verify $F = t^2$ (Corollary 7.8). (g) Using $t_{0.025,3} = 3.182$, give a 95% confidence interval for β_1 , and at $x_0 = 6$ a 95% confidence interval for the conditional mean and a 95% prediction interval for a new observation (Theorems 9.1 and 9.2).

Exercise D2. (Which assumption fails?) For each situation, state which of (A1)–(A5) is violated, and which of the following still hold: unbiasedness of $\hat{\beta}$ (Theorem 3.3), the variance formula $\text{var}(\hat{\beta}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ (Theorem 3.4), the Gauss–Markov property (Theorem 3.5), and the exact t and F tests of Section 7.

- (i) Household expenditure is regressed on income; the scatter of the observations around the fitted line widens visibly as income grows.

- (ii) Wages are regressed on years of schooling, an intercept, and indicator variables for *all four* regions of a country.
- (iii) Test scores are regressed on class size, but family income—which is correlated with class size and affects test scores—is not observed and is left out.
- (iv) Monthly sales are regressed on monthly advertising expenditure; the errors of consecutive months are positively correlated.
- (v) The response is the indicator $Y_i \in \{0, 1\}$ of whether individual i is employed, regressed on age and schooling.

Exercise D3. (A lower bound for leverages.) (a) Show that in a model with an intercept, $h_{ii} \geq 1/n$ for every i , so that together with Theorem 2.14, $1/n \leq h_{ii} \leq 1$. *Hint:* apply the argument of Theorem 2.14(i) to the symmetric idempotent matrix $\mathbf{H} - \mathbf{J}$ of Lemma 7.2, Step 2. (b) For the simple regression model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, show that $h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{l=1}^n (x_l - \bar{x})^2}$, and check this formula against the leverages found in Exercise D1(d).

Exercise D4. (Adding an orthogonal regressor.) Let $\mathbf{X}_{k+1} = [\mathbf{X}_k \ \mathbf{x}]$ with $\mathbf{X}_k' \mathbf{x} = 0$ and $\mathbf{x} \neq \mathbf{0}$. Show: (a) $\hat{\beta}_{k+1} = \mathbf{x}' \mathbf{Y} / (\mathbf{x}' \mathbf{x})$; (b) the coefficients on $\mathbf{X}_k$ are the same in the short regression (on $\mathbf{X}_k$) and in the enlarged regression (on $\mathbf{X}_{k+1}$); (c) $\text{SSE}_k - \text{SSE}_{k+1} = (\mathbf{x}' \mathbf{Y})^2 / (\mathbf{x}' \mathbf{x})$. (d) Relate (b) to Theorem 8.3.

12 Solutions to Exercises

Solutions to Tier A

Solution to Exercise A1. $\mathbf{a}' \boldsymbol{\beta} = \sum_i a_i \beta_i$ gives $\partial(\mathbf{a}' \boldsymbol{\beta}) / \partial \beta_j = a_j$, i.e. $D(\mathbf{a}' \boldsymbol{\beta}) = \mathbf{a}'$. In $\boldsymbol{\beta}' \mathbf{A} \boldsymbol{\beta} = \sum_{i,l} \beta_i A_{il} \beta_l$ the coordinate β_j occurs in the terms with $i = j$ and in those with $l = j$, so

$$\frac{\partial(\boldsymbol{\beta}' \mathbf{A} \boldsymbol{\beta})}{\partial \beta_j} = \sum_l A_{jl} \beta_l + \sum_i \beta_i A_{ij} = (\mathbf{A} \boldsymbol{\beta})_j + (\mathbf{A}' \boldsymbol{\beta})_j, \quad \text{i.e.} \quad D(\boldsymbol{\beta}' \mathbf{A} \boldsymbol{\beta}) = \boldsymbol{\beta}' (\mathbf{A} + \mathbf{A}')$$

(this is the proof of Lemma 2.2). Applying (i) with $\mathbf{a} = \mathbf{X}' \mathbf{Y}$ and (ii) with the symmetric $\mathbf{A} = \mathbf{X}' \mathbf{X}$ to $Q(\boldsymbol{\beta}) = \mathbf{Y}' \mathbf{Y} - 2\boldsymbol{\beta}' \mathbf{X}' \mathbf{Y} + \boldsymbol{\beta}' \mathbf{X}' \mathbf{X} \boldsymbol{\beta}$ gives $DQ(\boldsymbol{\beta}) = -2(\mathbf{X}' \mathbf{Y})' + 2\boldsymbol{\beta}' \mathbf{X}' \mathbf{X} = -2\mathbf{Y}' \mathbf{X} + 2\boldsymbol{\beta}' \mathbf{X}' \mathbf{X}$.

Solution to Exercise A2. (a) If $\mathbf{A} \mathbf{v} = \lambda \mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $\lambda \mathbf{v} = \mathbf{A} \mathbf{v} = \mathbf{A}^2 \mathbf{v} = \lambda^2 \mathbf{v}$, so $\lambda^2 = \lambda$ and $\lambda \in \{0, 1\}$. (b) $\text{tr}(\mathbf{A}) = \text{rank}(\mathbf{A})$ by Lemma 2.8, which holds for every idempotent matrix. For symmetric $\mathbf{A}$ the trace can also be read off the spectral decomposition: by (L3), $\mathbf{A} = \mathbf{P} \boldsymbol{\Lambda} \mathbf{P}'$ with $\mathbf{P}$ orthogonal and $\boldsymbol{\Lambda}$ diagonal with entries in $\{0, 1\}$ by (a), so $\text{tr}(\mathbf{A}) = \text{tr}(\boldsymbol{\Lambda} \mathbf{P}' \mathbf{P}) = \text{tr}(\boldsymbol{\Lambda})$ ((L2)) is the number of unit eigenvalues, and by the lemma this number is the rank. (c) $\mathbf{H}$ and $\mathbf{M}$ are symmetric idempotent (Theorem 2.9(i)) and $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$ has dimension $k + 1$ (Theorem 2.9(iv), Step 1), so $\text{tr}(\mathbf{H}) = \text{rank}(\mathbf{H}) = k + 1$ and $\text{tr}(\mathbf{M}) = \text{tr}(\mathbf{I}_n) - \text{tr}(\mathbf{H}) = n - k - 1$.

Solution to Exercise A3. Expand $Q(\boldsymbol{\beta}) = \mathbf{Y}' \mathbf{Y} - 2\boldsymbol{\beta}' \mathbf{X}' \mathbf{Y} + \boldsymbol{\beta}' \mathbf{X}' \mathbf{X} \boldsymbol{\beta}$ and differentiate with Lemma 2.2: $DQ(\boldsymbol{\beta}) = -2\mathbf{Y}' \mathbf{X} + 2\boldsymbol{\beta}' \mathbf{X}' \mathbf{X}$. Setting it to zero gives the normal equation $\mathbf{X}' \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}' \mathbf{Y}$. Under (A4), $\mathbf{X}' \mathbf{X} \mathbf{v} = \mathbf{0}$ implies $\|\mathbf{X} \mathbf{v}\|^2 = \mathbf{v}' \mathbf{X}' \mathbf{X} \mathbf{v} = 0$, so $\mathbf{X} \mathbf{v} = \mathbf{0}$ and $\mathbf{v} = \mathbf{0}$; hence $\mathbf{X}' \mathbf{X}$ is invertible and $\hat{\boldsymbol{\beta}} = (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \mathbf{Y}$. Second-order condition: the Hessian $2\mathbf{X}' \mathbf{X}$ satisfies $\mathbf{v}' (2\mathbf{X}' \mathbf{X}) \mathbf{v} = 2\|\mathbf{X} \mathbf{v}\|^2 > 0$ for $\mathbf{v} \neq \mathbf{0}$; the minimum is global and unique by $Q(\boldsymbol{\beta}) = Q(\hat{\boldsymbol{\beta}}) + \|\mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})\|^2$, equation (7).

Solution to Exercise A4. (a) $\mathbf{H}' = \mathbf{X}[(\mathbf{X}' \mathbf{X})^{-1}]' \mathbf{X}' = \mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' = \mathbf{H}$ since $(\mathbf{X}' \mathbf{X})^{-1}$ is symmetric; $\mathbf{H}^2 = \mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} (\mathbf{X}' \mathbf{X}) (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' = \mathbf{H}$; then $\mathbf{M}' = \mathbf{I} - \mathbf{H}' = \mathbf{M}$ and $\mathbf{M}^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H}^2 = \mathbf{M}$. (b) $\mathbf{H} \mathbf{X} = \mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \mathbf{X} = \mathbf{X}$, $\mathbf{M} \mathbf{X} = \mathbf{X} - \mathbf{H} \mathbf{X} = \mathbf{0}$, $\mathbf{H} \mathbf{M} = \mathbf{H} - \mathbf{H}^2 = \mathbf{0}$. (c) By Lemma 2.8, $\text{tr}(\mathbf{H}) = \text{rank}(\mathbf{H})$; every column of $\mathbf{H}$ is $\mathbf{X}$ times a vector, so $\mathcal{C}(\mathbf{H}) \subseteq \mathcal{C}(\mathbf{X})$, and $\mathbf{X} = \mathbf{H} \mathbf{X}$ gives the reverse inclusion; hence $\text{rank}(\mathbf{H}) = \dim \mathcal{C}(\mathbf{X}) = k + 1$ by (A4), and $\text{tr}(\mathbf{M}) = n - (k + 1)$.

Solution to Exercise A5. Substituting $\mathbf{Y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}$: $\hat{\boldsymbol{\beta}} = (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' (\mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta} + (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \boldsymbol{\varepsilon}$, where (A4) is what makes $(\mathbf{X}' \mathbf{X})^{-1}$ exist. Taking expectations and using $E[\boldsymbol{\varepsilon}] = \mathbf{0}$ (A2) gives

$E[\hat{\beta}] = \beta$. For the variance, $\hat{\beta}$ is the constant β plus the linear transformation $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon$, so Lemma 3.2(i) and (A3) give $\text{var}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.

Solution to Exercise A6. *Statement:* under (A1)–(A4), for every linear estimator $\tilde{\beta} = \mathbf{C}\mathbf{Y}$ that is unbiased for all β , $\text{var}(\tilde{\beta}) - \text{var}(\hat{\beta})$ is positive semi-definite, with equality iff $\tilde{\beta} = \hat{\beta}$. *Proof:* $E[\mathbf{C}\mathbf{Y}] = \mathbf{C}\mathbf{X}\beta = \beta$ for all β forces $\mathbf{C}\mathbf{X} = \mathbf{I}_{k+1}$. Write $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}$; then $\mathbf{D}\mathbf{X} = \mathbf{C}\mathbf{X} - \mathbf{I} = \mathbf{0}$. By Lemma 3.2 and (A3), $\text{var}(\tilde{\beta}) = \sigma^2\mathbf{C}\mathbf{C}' = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} + \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}' + \sigma^2\mathbf{D}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \sigma^2\mathbf{D}\mathbf{D}'$, and the two middle terms vanish because $\mathbf{D}\mathbf{X} = \mathbf{0}$ and $\mathbf{X}'\mathbf{D}' = (\mathbf{D}\mathbf{X})'$. Hence $\text{var}(\tilde{\beta}) - \text{var}(\hat{\beta}) = \sigma^2\mathbf{D}\mathbf{D}' \succeq \mathbf{0}$, with equality iff $\mathbf{D} = \mathbf{0}$, i.e. $\tilde{\beta} = \hat{\beta}$.

Solution to Exercise A7. $\mathbf{e} = \mathbf{M}\mathbf{Y} = \mathbf{M}(\mathbf{X}\beta + \varepsilon) = \mathbf{M}\varepsilon$ since $\mathbf{M}\mathbf{X} = \mathbf{0}$, so $\text{SSE} = \varepsilon'\mathbf{M}'\mathbf{M}\varepsilon = \varepsilon'\mathbf{M}\varepsilon$ ($\mathbf{M}$ symmetric idempotent). By Lemma 3.6 with $E[\varepsilon] = \mathbf{0}$ (A2) and $\text{var}(\varepsilon) = \sigma^2\mathbf{I}$ (A3), $E[\text{SSE}] = \text{tr}(\mathbf{M} \cdot \sigma^2\mathbf{I}) = \sigma^2 \text{tr}(\mathbf{M}) = \sigma^2(n - k - 1)$ (Theorem 2.9(iv)). Dividing by $n - k - 1$ gives $E[\text{MSE}] = \sigma^2$.

Solution to Exercise A8. (a) $M_{\mathbf{A}\mathbf{z}+\mathbf{b}}(\mathbf{t}) = E[e^{\mathbf{t}'(\mathbf{A}\mathbf{z}+\mathbf{b})}] = e^{\mathbf{t}'\mathbf{b}}M_{\mathbf{z}}(\mathbf{A}'\mathbf{t}) = \exp(\mathbf{t}'(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}) + \frac{1}{2}\mathbf{t}'\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'\mathbf{t})$, the MGF of $N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$; conclude by uniqueness (N1). (b) Write $\mathbf{A} = \mathbf{P}\mathbf{A}\mathbf{P}'$ ((L3)) with $\mathbf{A} = \text{diag}(1, \dots, 1, 0, \dots, 0)$ (r ones: the eigenvalues lie in $\{0, 1\}$ by idempotency and $\text{tr } \mathbf{A} = \text{rank } \mathbf{A} = r$). Then $\mathbf{w} = \mathbf{P}'\mathbf{z} \sim N(\mathbf{0}, \mathbf{P}'\mathbf{P}) = N(\mathbf{0}, \mathbf{I}_n)$ by (a), its coordinates are i.i.d. $N(0, 1)$ (the MGF $e^{\mathbf{t}'\mathbf{t}/2}$ factorizes, (N2)), and $\mathbf{z}'\mathbf{A}\mathbf{z} = \mathbf{w}'\mathbf{A}\mathbf{w} = \sum_{i=1}^r w_i^2 \sim \chi_r^2$ by definition (N5).

Solution to Exercise A9. $(\mathbf{U}', \mathbf{V}')' = ((\mathbf{A}_1\mathbf{z})', (\mathbf{A}_2\mathbf{z})')'$ is a linear image of $\mathbf{z}$, hence jointly normal (Theorem 4.2), with $\text{cov}(\mathbf{U}, \mathbf{V}) = \mathbf{A}_1\mathbf{I}\mathbf{A}_2' = \mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$ (Lemma 3.2(ii)). By Theorem 4.3, $\mathbf{U} \perp\!\!\!\perp \mathbf{V}$. Since $\mathbf{A}_i$ is symmetric idempotent, $\mathbf{z}'\mathbf{A}_i\mathbf{z} = \|\mathbf{A}_i\mathbf{z}\|^2$ is a function of $\mathbf{U}$ (resp. $\mathbf{V}$) alone, so the two quadratic forms are independent ((N3)).

Solution to Exercise A10. Let $\mathbf{z} = \varepsilon/\sigma \sim N(\mathbf{0}, \mathbf{I}_n)$ (Theorem 4.2). Then $\text{SSE}/\sigma^2 = \mathbf{z}'\mathbf{M}\mathbf{z}$ with $\mathbf{M}$ symmetric idempotent of trace $n - k - 1$, so $\text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ by Theorem 4.5. For independence, stack $\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon$ and $\mathbf{e} = \mathbf{M}\varepsilon$: the pair is jointly normal by Theorem 4.2, with cross-covariance $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} = \mathbf{0}$ because $\mathbf{X}'\mathbf{M} = (\mathbf{M}\mathbf{X})' = \mathbf{0}$. Theorem 4.3 gives $\hat{\beta} \perp\!\!\!\perp \mathbf{e}$, hence $\hat{\beta} \perp\!\!\!\perp \text{SSE} = \mathbf{e}'\mathbf{e}$ ((N3)).

Solution to Exercise A11. *Decomposition:* $\mathbf{M}_0\mathbf{Y} = \mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{M}_0\mathbf{e} = \mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{e}$ because $\mathbf{1}'\mathbf{e} = 0$ (Theorem 2.12(ii)); the cross term $\mathbf{e}'\mathbf{M}_0\hat{\mathbf{Y}} = \mathbf{e}'\hat{\mathbf{Y}} - \bar{Y}\mathbf{e}'\mathbf{1} = 0$ (Theorem 2.12(ii), (iii)), so $\text{SST} = \|\mathbf{M}_0\mathbf{Y}\|^2 = \|\mathbf{M}_0\hat{\mathbf{Y}}\|^2 + \|\mathbf{e}\|^2 = \text{SSR} + \text{SSE}$ (using $\bar{\hat{Y}} = \bar{Y}$). *F statistic:* $\hat{\mathbf{Y}} - \bar{Y}\mathbf{1} = (\mathbf{H} - \mathbf{J})\mathbf{Y}$, and $\mathbf{H} - \mathbf{J}$ is symmetric idempotent of rank k because $\mathbf{H}\mathbf{J} = \mathbf{J} = \mathbf{J}\mathbf{H}$ (from $\mathbf{H}\mathbf{1} = \mathbf{1}$). Under H_0 , $\mathbf{X}\beta = \beta_0\mathbf{1}$ and $(\mathbf{H} - \mathbf{J})\mathbf{1} = \mathbf{0}$, so $\text{SSR} = \varepsilon'(\mathbf{H} - \mathbf{J})\varepsilon$ and $\text{SSR}/\sigma^2 \sim \chi_k^2$ (Theorem 4.5); $(\mathbf{H} - \mathbf{J})\mathbf{M} = \mathbf{0}$ gives $\text{SSR} \perp\!\!\!\perp \text{SSE}$ (Theorem 4.6); with $\text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ (Theorem 5.2), $F = (\text{SSR}/k)/(\text{SSE}/(n - k - 1)) \sim F_{k, n-k-1}$ by definition (N5).

Solutions to Tier B

Solution to Exercise B1. (a) $\mathbf{H} = \mathbf{H}'\mathbf{H}$ gives $h_{ii} = \sum_j h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2$, so $h_{ii} \geq 0$ and $h_{ii} \geq h_{ii}^2$, i.e. $0 \leq h_{ii} \leq 1$. (b) $\sum_i h_{ii} = \text{tr}(\mathbf{H}) = \text{rank}(\mathbf{H}) = k + 1$ (Lemma 2.8, Theorem 2.9(iv)). (c) $\mathbf{e} = \mathbf{M}\varepsilon$, so $\text{var}(\mathbf{e}) = \mathbf{M}(\sigma^2\mathbf{I})\mathbf{M}' = \sigma^2\mathbf{M}$ (Lemma 3.2, A3), whose (i, i) entry is $\sigma^2(1 - h_{ii})$.

Solution to Exercise B2. (a) By Theorem 4.2, $\mathbf{c}'\hat{\beta} \sim N(\mathbf{c}'\beta, \sigma^2\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c})$. (b) Under H_0 , $Z = (\mathbf{c}'\hat{\beta} - r_0)/(\sigma\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}) \sim N(0, 1)$ by (a); $V = \text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ is independent of $\hat{\beta}$ (Theorem 5.2), hence of Z ((N3)); so $t = Z/\sqrt{V/(n - k - 1)} = (\mathbf{c}'\hat{\beta} - r_0)/(s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}) \sim t_{n-k-1}$ ((N5)). Squaring, $t^2 = (Z^2/1)/(V/(n - k - 1))$ with $Z^2 \sim \chi_1^2$ (definition (N5) with $r = 1$) independent of V , so $t^2 \sim F_{1, n-k-1}$; it coincides with the F statistic of Theorem 7.4 for $\mathbf{R} = \mathbf{c}'$, $r = r_0$ (Corollary 7.8).

Solution to Exercise B3. Let $\mathbf{X}_U$ be the full design matrix and $\mathbf{X}_R$ the submatrix of its first $k_0 + 1$ columns (intercept and the k_0 retained regressors), with hat matrices $\mathbf{H}_U = \mathbf{H}$, $\mathbf{H}_R = \mathbf{X}_R(\mathbf{X}_R'\mathbf{X}_R)^{-1}\mathbf{X}_R'$ and residual makers $\mathbf{M}_U = \mathbf{M}$, $\mathbf{M}_R = \mathbf{I} - \mathbf{H}_R$; the reduced model is the full model under $H_0: \beta_{k_0+1} = \dots = \beta_k = 0$.

(a) The reduced model minimizes $\|\mathbf{Y} - \mathbf{X}_U \boldsymbol{\beta}\|^2$ over the subspace $\{\boldsymbol{\beta} : \beta_{k_0+1} = \dots = \beta_k = 0\}$ of $\mathbb{R}^{k+1}$, whereas the unconstrained minimum SSE_U is taken over all of $\mathbb{R}^{k+1}$; hence $\text{SSE}_R \geq \text{SSE}_U$. (Alternatively, by (b) below, $\text{SSE}_R - \text{SSE}_U = \|(\mathbf{H}_U - \mathbf{H}_R)\mathbf{Y}\|^2 \geq 0$.)

(b) *Nested projections.* Each column of $\mathbf{X}_R$ is a column of $\mathbf{X}_U$, so $\mathbf{H}_U \mathbf{X}_R = \mathbf{X}_R$ (Theorem 2.9(ii)); hence $\mathbf{H}_U \mathbf{H}_R = \mathbf{H}_U \mathbf{X}_R (\mathbf{X}_R' \mathbf{X}_R)^{-1} \mathbf{X}_R' = \mathbf{H}_R$ and, transposing, $\mathbf{H}_R \mathbf{H}_U = \mathbf{H}_R$. Therefore $\mathbf{M}_R - \mathbf{M}_U = \mathbf{H}_U - \mathbf{H}_R$ is symmetric, idempotent since $(\mathbf{H}_U - \mathbf{H}_R)^2 = \mathbf{H}_U - \mathbf{H}_R - \mathbf{H}_R + \mathbf{H}_R = \mathbf{H}_U - \mathbf{H}_R$, of rank $\text{tr}(\mathbf{H}_U) - \text{tr}(\mathbf{H}_R) = (k+1) - (k_0+1) = k - k_0$ (Lemma 2.8), and $(\mathbf{M}_R - \mathbf{M}_U)\mathbf{M}_U = (\mathbf{H}_U - \mathbf{H}_R)(\mathbf{I} - \mathbf{H}_U) = \mathbf{H}_U - \mathbf{H}_U - \mathbf{H}_R + \mathbf{H}_R = \mathbf{0}$. *Distribution.* $\text{SSE}_R - \text{SSE}_U = \mathbf{Y}'\mathbf{M}_R\mathbf{Y} - \mathbf{Y}'\mathbf{M}_U\mathbf{Y} = \mathbf{Y}'(\mathbf{M}_R - \mathbf{M}_U)\mathbf{Y}$. Under H_0 , $\mathbf{Y} = \mathbf{X}_R \boldsymbol{\beta}_R + \boldsymbol{\varepsilon}$ (with $\boldsymbol{\beta}_R$ the first $k_0 + 1$ coefficients) and $\mathbf{M}_R \mathbf{X}_R = \mathbf{0} = \mathbf{M}_U \mathbf{X}_R$, so $\text{SSE}_R - \text{SSE}_U = \boldsymbol{\varepsilon}'(\mathbf{M}_R - \mathbf{M}_U)\boldsymbol{\varepsilon}$ and likewise $\text{SSE}_U = \boldsymbol{\varepsilon}'\mathbf{M}_U\boldsymbol{\varepsilon}$. With $\mathbf{z} = \boldsymbol{\varepsilon}/\sigma \sim N(\mathbf{0}, \mathbf{I}_n)$, Theorem 4.5 gives $(\text{SSE}_R - \text{SSE}_U)/\sigma^2 \sim \chi_{k-k_0}^2$, and Theorem 4.6 (with $\mathbf{A}_1 = \mathbf{M}_R - \mathbf{M}_U$, $\mathbf{A}_2 = \mathbf{M}_U$, $\mathbf{A}_1 \mathbf{A}_2 = \mathbf{0}$) gives independence from $\text{SSE}_U/\sigma^2 \sim \chi_{n-k-1}^2$. Hence $F = \frac{(\text{SSE}_R - \text{SSE}_U)/(k-k_0)}{\text{SSE}_U/(n-k-1)} \sim F_{k-k_0, n-k-1}$ by definition (N5). *Via Theorem 7.4.* The same F is the general F statistic for $\mathbf{R} = [\mathbf{0} \mid \mathbf{I}_{k-k_0}]$ ($q = k - k_0$) and $\mathbf{r} = \mathbf{0}$: by Corollary 7.6, its numerator $(\mathbf{R}\hat{\boldsymbol{\beta}})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}]^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}})/q$ equals $(\text{SSE}_R - \text{SSE}_U)/q$.

Solution to Exercise B4. By Theorems 5.1 and 4.2, $\mathbf{R}\hat{\boldsymbol{\beta}} \sim N(\mathbf{R}\boldsymbol{\beta}, \sigma^2 \boldsymbol{\Sigma}_R)$ with $\boldsymbol{\Sigma}_R = \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$, which is positive definite because $\text{rank } \mathbf{R} = q$ (for $\mathbf{v} \neq \mathbf{0}$, $\mathbf{R}'\mathbf{v} \neq \mathbf{0}$ and $(\mathbf{X}'\mathbf{X})^{-1}$ is positive definite by Theorem 2.4, Step 4). Under H_0 , $\mathbf{w} = \sigma^{-1} \boldsymbol{\Sigma}_R^{-1/2}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) \sim N(\mathbf{0}, \mathbf{I}_q)$ ((L4), Theorem 4.2), so $W = \mathbf{w}'\mathbf{w} = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\boldsymbol{\Sigma}_R^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})/\sigma^2 \sim \chi_q^2$ (Theorem 4.5 with $\mathbf{A} = \mathbf{I}_q$). W is a function of $\hat{\boldsymbol{\beta}}$, hence independent of $V = \text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ (Theorem 5.2, (N3)), and $F = (W/q)/(V/(n-k-1))$ is the displayed statistic (the factors σ^{-2} cancel), distributed as $F_{q, n-k-1}$ by (N5).

Solution to Exercise B5. (a) Unbiasedness: $E[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\boldsymbol{\varepsilon}] = \boldsymbol{\beta}$ (unchanged; unbiasedness does not use (A3)). Actual variance:

$$\text{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\boldsymbol{\Omega})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1},$$

which in general differs from $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$, so the usual standard errors are wrong. (Not for *every* $\boldsymbol{\Omega} \neq \mathbf{I}$, however: if $\boldsymbol{\Omega} = \mathbf{I} + \mathbf{v}\mathbf{v}'$ with $\mathbf{X}'\mathbf{v} = \mathbf{0}$, then $\mathbf{X}'\boldsymbol{\Omega}\mathbf{X} = \mathbf{X}'\mathbf{X}$ and the two expressions coincide.)

(b) Let $\boldsymbol{\Omega} = \mathbf{L}\mathbf{L}'$ with $\mathbf{L}$ invertible ((L4); for instance the Cholesky factor, or the symmetric square root $\boldsymbol{\Omega}^{1/2}$). Transform: $\mathbf{L}^{-1}\mathbf{Y} = \mathbf{L}^{-1}\mathbf{X}\boldsymbol{\beta} + \mathbf{L}^{-1}\boldsymbol{\varepsilon}$, where $\text{var}(\mathbf{L}^{-1}\boldsymbol{\varepsilon}) = \sigma^2\mathbf{L}^{-1}\boldsymbol{\Omega}(\mathbf{L}^{-1})' = \sigma^2\mathbf{I}$. The OLS estimator applied to the transformed model is

$$\hat{\boldsymbol{\beta}}_{\text{GLS}} = [(\mathbf{L}^{-1}\mathbf{X})'(\mathbf{L}^{-1}\mathbf{X})]^{-1}(\mathbf{L}^{-1}\mathbf{X})'(\mathbf{L}^{-1}\mathbf{Y}) = (\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{Y}.$$

Since the transformed model satisfies (A1)–(A4) (its regressor matrix $\mathbf{L}^{-1}\mathbf{X}$ has full column rank because $\mathbf{L}^{-1}$ is invertible), Gauss–Markov (Theorem 3.5) guarantees that $\hat{\boldsymbol{\beta}}_{\text{GLS}}$ is BLUE in the transformed model. It is also BLUE in the original model, because the two models have the *same* class of linear unbiased estimators with the *same* variances: an estimator $\mathbf{C}\mathbf{Y}$ that is linear in $\mathbf{Y}$ equals $(\mathbf{C}\mathbf{L})(\mathbf{L}^{-1}\mathbf{Y})$, which is linear in the transformed response, and conversely every $\tilde{\mathbf{C}}(\mathbf{L}^{-1}\mathbf{Y})$ equals $(\tilde{\mathbf{C}}\mathbf{L}^{-1})\mathbf{Y}$; unbiasedness in the original model, $\mathbf{C}\mathbf{X} = \mathbf{I}$, is the same condition as $(\mathbf{C}\mathbf{L})(\mathbf{L}^{-1}\mathbf{X}) = \mathbf{I}$ in the transformed model; and $\text{var}(\mathbf{C}\mathbf{Y})$ is the variance of the same random vector under both descriptions. Hence the estimator that minimizes the variance (in the positive semi-definite order) over the transformed class, namely $\hat{\boldsymbol{\beta}}_{\text{GLS}}$, minimizes it over the original class.

Solution to Exercise B6. (a) $\hat{Y}_0 - E[Y_0] = \mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$, so by Lemma 3.2 and Theorem 3.4, $\text{var}(\hat{Y}_0 - E[Y_0]) = \mathbf{x}_0' \text{var}(\hat{\boldsymbol{\beta}})\mathbf{x}_0 = \sigma^2 \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$. (b) $\hat{Y}_0 - Y_0 = \mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - \varepsilon_0$, where ε_0 is independent of $\boldsymbol{\varepsilon}$ and hence of $\mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$; the variance of a sum of independent terms is the sum of the variances (fact (N7)), so $\text{var}(\hat{Y}_0 - Y_0) = \sigma^2 \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 + \sigma^2 = \sigma^2(1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)$ (Theorem 9.2, Step 2). (c) The prediction variance exceeds the estimation variance by exactly σ^2 , the irreducible noise variance of ε_0 .

Solutions to Tier C

Solution to Exercise C1. (a) The second block of the normal equations $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}$ reads $\mathbf{X}'_2\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}'_2\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \mathbf{X}'_2\mathbf{Y}$, so $\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}'_2\mathbf{X}_2)^{-1}\mathbf{X}'_2(\mathbf{Y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1)$. Substituting into the first block, $\mathbf{X}'_1\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}'_1\mathbf{H}_2(\mathbf{Y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1) = \mathbf{X}'_1\mathbf{Y}$ with $\mathbf{H}_2 = \mathbf{I} - \mathbf{M}_2$, which simplifies to $\mathbf{X}'_1\mathbf{M}_2\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 = \mathbf{X}'_1\mathbf{M}_2\mathbf{Y}$; here $\mathbf{X}'_1\mathbf{M}_2\mathbf{X}_1 = (\mathbf{M}_2\mathbf{X}_1)'(\mathbf{M}_2\mathbf{X}_1)$ is positive definite because $\mathbf{M}_2\mathbf{X}_1\mathbf{v} = \mathbf{0}$ would put $\mathbf{X}_1\mathbf{v}$ in $\mathcal{C}(\mathbf{X}_2)$, contradicting (A4) unless $\mathbf{v} = \mathbf{0}$. (b) The residual from regressing $\mathbf{M}_2\mathbf{Y}$ on $\mathbf{M}_2\mathbf{X}_1$ is $\mathbf{M}_2\mathbf{Y} - \mathbf{M}_2\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 = \mathbf{M}_2(\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 + \mathbf{e}) = \mathbf{M}_2\mathbf{e} = \mathbf{e}$, using $\mathbf{M}_2\mathbf{X}_2 = \mathbf{0}$ and $\mathbf{X}'_2\mathbf{e} = \mathbf{0}$ (Theorem 2.12(i)).

Solution to Exercise C2. Substituting the true model into the short-regression estimator, $\tilde{\boldsymbol{\gamma}} = (\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}) = \boldsymbol{\beta}_1 + (\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{X}_2\boldsymbol{\beta}_2 + (\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1\boldsymbol{\varepsilon}$; taking expectations with $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ (A2) gives the formula. Unbiasedness holds iff $\mathbf{X}'_1\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{0}$, which occurs when $\mathbf{X}'_1\mathbf{X}_2 = \mathbf{0}$ (every column of $\mathbf{X}_1$ is orthogonal to every column of $\mathbf{X}_2$) or $\boldsymbol{\beta}_2 = \mathbf{0}$.

Solution to Exercise C3. (a) With $\mathcal{C}_k \subseteq \mathcal{C}_{k+1}$ the two column spaces, $\hat{\mathbf{Y}}_{k+1}$ is the closest point of $\mathcal{C}_{k+1}$ to $\mathbf{Y}$ (Theorem 2.5) and $\hat{\mathbf{Y}}_k \in \mathcal{C}_k \subseteq \mathcal{C}_{k+1}$, so $\text{SSE}_{k+1} = \|\mathbf{Y} - \hat{\mathbf{Y}}_{k+1}\|^2 \leq \|\mathbf{Y} - \hat{\mathbf{Y}}_k\|^2 = \text{SSE}_k$; since SST is the same in both models, $R^2 = 1 - \text{SSE}/\text{SST}$ cannot decrease. (b) $\bar{R}_{k+1}^2 > \bar{R}_k^2 \iff \text{MSE}_{k+1} < \text{MSE}_k \iff \text{SSE}_{k+1}/(n - k - 2) < (\text{SSE}_{k+1} + \hat{\boldsymbol{\beta}}_{k+1}'\mathbf{x}'\mathbf{M}_k\mathbf{x})/(n - k - 1)$ (Lemma 6.9) $\iff \text{SSE}_{k+1} < (n - k - 2)\hat{\boldsymbol{\beta}}_{k+1}'\mathbf{x}'\mathbf{M}_k\mathbf{x} \iff 1 < \hat{\boldsymbol{\beta}}_{k+1}'\mathbf{x}'\mathbf{M}_k\mathbf{x}/\text{MSE}_{k+1}$; the right-hand side is t_{k+1}^2 because $\text{SE}(\hat{\boldsymbol{\beta}}_{k+1})^2 = \text{MSE}_{k+1}[(\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}]_{k+1,k+1} = \text{MSE}_{k+1}/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$ by (31).

Solution to Exercise C4. (a) With $\mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\lambda}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + 2\boldsymbol{\lambda}'(\mathbf{R}\boldsymbol{\beta} - \mathbf{r})$, Lemma 2.2 gives $\partial\mathcal{L}/\partial\boldsymbol{\beta}' = -2\mathbf{Y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X} + 2\boldsymbol{\lambda}'\mathbf{R} = \mathbf{0}'$, i.e. $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}}_R = \mathbf{X}'\mathbf{Y} - \mathbf{R}'\hat{\boldsymbol{\lambda}}$, so $\hat{\boldsymbol{\beta}}_R = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}}$; imposing $\mathbf{R}\hat{\boldsymbol{\beta}}_R = \mathbf{r}$ gives $\hat{\boldsymbol{\lambda}} = [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})$ ($\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$ is positive definite since $\text{rank } \mathbf{R} = q$). (b) Substitute $\hat{\boldsymbol{\lambda}}$ into the expression for $\hat{\boldsymbol{\beta}}_R$. (c) For any $\boldsymbol{\beta}$ with $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$, $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R\|^2 + \|\mathbf{X}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta})\|^2$, because the cross term is $(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R)'\mathbf{X}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta}) = \hat{\boldsymbol{\lambda}}'\mathbf{R}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta}) = 0$ (Theorem 10.3, Step 3); so $\hat{\boldsymbol{\beta}}_R$ is the constrained minimizer, and $\text{SSE}_R \geq \text{SSE}_U$ because the unconstrained minimum SSE_U is taken over a larger set.

Solutions to Tier D

Solution to Exercise D1. (a) $\mathbf{X}'\mathbf{X} = \begin{pmatrix} 5 & 15 \\ 15 & 55 \end{pmatrix}$, $\mathbf{X}'\mathbf{Y} = (15, 57)'$, $\det(\mathbf{X}'\mathbf{X}) = 275 - 225 = 50$, $(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{50} \begin{pmatrix} 55 & -15 \\ -15 & 5 \end{pmatrix} = \begin{pmatrix} 1.1 & -0.3 \\ -0.3 & 0.1 \end{pmatrix}$, and $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = (16.5 - 17.1, -4.5 + 5.7)' = (-0.6, 1.2)'$. (b) $\hat{Y}_i = -0.6 + 1.2x_i$: $\hat{\mathbf{Y}} = (0.6, 1.8, 3.0, 4.2, 5.4)'$ and $\mathbf{e} = (0.4, 0.2, -1.0, -0.2, 0.6)'$; $\sum e_i = 0$ and $\sum x_i e_i = 0.4 + 0.4 - 3.0 - 0.8 + 3.0 = 0$. (c) $\text{SSE} = 0.16 + 0.04 + 1 + 0.04 + 0.36 = 1.6$, $\text{MSE} = 1.6/3 = 0.5333$, $s = 0.7303$. (d) $h_{ii} = \mathbf{x}'_i(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i = 1.1 - 0.6x_i + 0.1x_i^2$, giving $(0.6, 0.3, 0.2, 0.3, 0.6)$: all in $[0, 1]$ and summing to $2 = k + 1$. (e) $\bar{Y} = 3$, $\text{SST} = 4 + 1 + 1 + 1 + 9 = 16$, $\text{SSR} = \sum (\hat{Y}_i - 3)^2 = 5.76 + 1.44 + 0 + 1.44 + 5.76 = 14.4$, and indeed $14.4 + 1.6 = 16$; $R^2 = 14.4/16 = 0.9$ and $\bar{R}^2 = 1 - (1.6/3)/(16/4) = 0.8667$. (f) $\text{SE}(\hat{\beta}_1) = s\sqrt{0.1} = \sqrt{0.05333} = 0.2309$, $t = 1.2/0.2309 = 5.196$; $F = \text{MSR}/\text{MSE} = 14.4/0.5333 = 27.0 = t^2$. (g) For β_1 : $1.2 \pm 3.182 \times 0.2309 = 1.2 \pm 0.735$, i.e. $(0.465, 1.935)$. At $\mathbf{x}_0 = (1, 6)'$: $\mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 1.1 - 3.6 + 3.6 = 1.1$ and $\hat{Y}_0 = -0.6 + 7.2 = 6.6$; the confidence interval for the conditional mean is $6.6 \pm 3.182 \times 0.7303\sqrt{1.1} = 6.6 \pm 2.437$, i.e. $(4.163, 9.037)$, and the prediction interval is $6.6 \pm 3.182 \times 0.7303\sqrt{2.1} = 6.6 \pm 3.368$, i.e. $(3.232, 9.968)$.

Solution to Exercise D2. (i) Heteroscedasticity: (A3) fails ($\text{var}(\varepsilon_i)$ depends on income) while (A1)/(A2) may well hold. Unbiasedness survives (Theorem 3.3 uses only (A2) and (A4)); the variance formula, the Gauss–Markov property, $\mathbb{E}[\text{MSE}] = \sigma^2$ and the exact t and F tests all rely on (A3) and fail (compare Exercise B5). (ii) The dummy variable trap: (A4) fails, because the four indicators sum to the intercept column, $\mathbf{1} = D_1 + D_2 + D_3 + D_4$; $\mathbf{X}'\mathbf{X}$ is singular and $\hat{\boldsymbol{\beta}}$ is not defined. Drop one indicator (or the intercept). (iii) Omitted variable: the unobserved income is absorbed into the error, which is then correlated with class size, so (A2)—equivalently (A1)—fails for the regression that is estimated. Taking the four properties in turn: unbiasedness fails, the estimated coefficient vector being the short-regression estimator $\tilde{\boldsymbol{\gamma}}$ of Theorem 8.3, with $\mathbb{E}[\tilde{\boldsymbol{\gamma}}] = \boldsymbol{\beta}_1 + (\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{X}_2\boldsymbol{\beta}_2 \neq \boldsymbol{\beta}_1$ in general; the variance formula survives as a statement about dispersion, since if the true error $\boldsymbol{\varepsilon}$ satisfies

(A3) then Lemma 3.2 gives $\text{var}(\tilde{\gamma}) = \sigma^2(\mathbf{X}'_1\mathbf{X}_1)^{-1}$, but it is dispersion around the wrong centre; the Gauss–Markov property does not apply, because $\tilde{\gamma}$ is not in the class of unbiased estimators that Theorem 3.5 compares; and the exact t and F tests fail, because under (A5) the statistics are still built from a normal vector, but one centred at $\beta_1 + (\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{X}_2\beta_2$ rather than at β_1 , so the null distributions of Section 7 no longer hold. (iv) Serial correlation: (A3) fails through $\text{cov}(\varepsilon_t, \varepsilon_{t+1}) \neq 0$; the consequences are as in (i)—unbiased $\hat{\beta}$, wrong variance formula, invalid tests. (v) A binary response: (A5) fails, since given $\mathbf{x}_i$ the error $\varepsilon_i = Y_i - \mathbf{x}'_i\beta$ takes only the two values $1 - \mathbf{x}'_i\beta$ and $-\mathbf{x}'_i\beta$; (A3) also fails, since $\text{var}(\varepsilon_i) = p_i(1 - p_i)$ with $p_i = \mathbf{x}'_i\beta$ varies with i (and (A1) itself, which requires $p_i = \mathbf{x}'_i\beta \in [0, 1]$ for all i , is questionable). Taking the four properties in turn: if (A1) holds, $\hat{\beta}$ remains unbiased (Theorem 3.3 needs only (A2) and (A4)); the variance formula fails, since Lemma 3.2 gives $\text{var}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\text{var}(\varepsilon)\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$ with $\text{var}(\varepsilon)$ having diagonal entries $p_i(1 - p_i)$ that vary with i , so $\text{var}(\varepsilon) \neq \sigma^2\mathbf{I}$; the Gauss–Markov property fails together with (A3), exactly as in (i); and the exact t and F tests fail, since both (A3) and (A5) are violated and the normal-theory distributions of Sections 5–7 are unavailable.

Solution to Exercise D3. (a) By Lemma 7.2, Step 2, $\mathbf{H} - \mathbf{J}$ is symmetric idempotent whenever $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, so $\mathbf{H} - \mathbf{J} = (\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J})$ and, comparing (i, i) entries exactly as in the proof of Theorem 2.14(i), $h_{ii} - \frac{1}{n} = \sum_{j=1}^n (h_{ij} - \frac{1}{n})^2 \geq 0$. (b) For $\mathbf{X} = [\mathbf{1} \ \mathbf{x}]$, $\mathbf{X}'\mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_l x_l^2 \end{pmatrix}$ has determinant $n \sum_l x_l^2 - n^2\bar{x}^2 = nS_{xx}$, where $S_{xx} := \sum_l (x_l - \bar{x})^2 = \sum_l x_l^2 - n\bar{x}^2$; so $(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum_l x_l^2 & -n\bar{x} \\ -n\bar{x} & n \end{pmatrix}$ and

$$h_{ii} = (1, x_i) (\mathbf{X}'\mathbf{X})^{-1} (1, x_i)' = \frac{\sum_l x_l^2 - 2n\bar{x}x_i + nx_i^2}{nS_{xx}} = \frac{S_{xx} + n(x_i - \bar{x})^2}{nS_{xx}} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}.$$

In Exercise D1, $\bar{x} = 3$ and $S_{xx} = 10$, giving $h_{ii} = 0.2 + (x_i - 3)^2/10 = (0.6, 0.3, 0.2, 0.3, 0.6)$, as found there.

Solution to Exercise D4. $\mathbf{X}'_k\mathbf{x} = \mathbf{0}$ gives $\mathbf{H}_k\mathbf{x} = \mathbf{X}_k(\mathbf{X}'_k\mathbf{X}_k)^{-1}\mathbf{X}'_k\mathbf{x} = \mathbf{0}$, so $\mathbf{M}_k\mathbf{x} = \mathbf{x}$, $\mathbf{x}'\mathbf{M}_k\mathbf{x} = \mathbf{x}'\mathbf{x} > 0$ (hence Lemma 6.9 applies) and $\mathbf{x}'\mathbf{M}_k\mathbf{Y} = \mathbf{x}'\mathbf{Y}$. (a) Equation (27): $\hat{\beta}_{k+1} = \mathbf{x}'\mathbf{Y}/(\mathbf{x}'\mathbf{x})$. (b) In the proof of Lemma 6.9, $\hat{\beta}_k^* = \hat{\beta}_k - (\mathbf{X}'_k\mathbf{X}_k)^{-1}\mathbf{X}'_k\mathbf{x}\hat{\beta}_{k+1} = \hat{\beta}_k$ because $\mathbf{X}'_k\mathbf{x} = \mathbf{0}$. (c) Equation (28): $\text{SSE}_k - \text{SSE}_{k+1} = (\mathbf{x}'\mathbf{M}_k\mathbf{Y})^2/(\mathbf{x}'\mathbf{M}_k\mathbf{x}) = (\mathbf{x}'\mathbf{Y})^2/(\mathbf{x}'\mathbf{x})$. (d) Read the enlarged model as the true model and the short regression as the one omitting $\mathbf{x}$: Theorem 8.3 with $\mathbf{X}_1 = \mathbf{X}_k$ and $\mathbf{X}_2 = \mathbf{x}$ says that the omitted-variable bias is $(\mathbf{X}'_k\mathbf{X}_k)^{-1}\mathbf{X}'_k\mathbf{x}\beta_{k+1} = \mathbf{0}$; part (b) is the exact, sample-level counterpart of this statement about expectations.